

Generic Identifiability of Strongly Tree-Child Level-2

Jukes–Cantor Networks

A sharp boundary at weak tree-childness

Alec Kriebel

Independent Researcher

Correspondence: me@aleckriebel.com

August 2026

Abstract

Phylogenetic networks extend trees by allowing reticulation events, but a topology is scientifically meaningful only to the extent that it is identifiable from the distribution induced at the observed taxa. We study the displayed-tree Jukes–Cantor model on binary standard semi-directed level-2 networks. A semi-directed topology is called strongly tree-child here when every admissible rooting is tree-child. We first prove a structural fact: any binary level-2 semi-directed topology that admits even one tree-child rooting automatically has at most one triangle in each blob. We then show that the exact generic leaf-pattern distribution determines every strongly tree-child level-2 topology up to ordinary triangle redirection, the only topology change that can remain observationally compatible on a full-dimensional model region. More precisely, one-sided full-dimensional stochastic containment occurs exactly when the labelled bridge trees agree and corresponding blobs are isomorphic or triangle-redirected. The proof combines Fourier cut factorization, pointwise cut-split recovery, projective extraction of local boundary tensors, bounded-support reconstruction of cycle and theta blobs, and exact finite atlases. The final atlas case consists of 192 seven-outgoing-port cycle-to-theta completions; established Jukes–Cantor trinet inequalities, applied to restored support labels, separate all of them. Sharpness holds at weak tree-childness: for every number of leaves at least four, two nonisomorphic and non-triangle-equivalent weakly but not strongly tree-child networks share a full-dimensional regular Jukes–Cantor model region. Exact graph, polynomial, sign, and Jacobian certificates accompany the proof.

Keywords. phylogenetic networks; identifiability; Jukes–Cantor model; level-2 networks; tree-child networks; algebraic statistics; semialgebraic geometry.

MSC 2020. 92B10, 62R01, 14P10, 05C05.

1 Introduction

Reticulate evolutionary events—hybridization, introgression, recombination, and lateral gene transfer—cannot in general be represented by a phylogenetic tree. A phylogenetic network replaces the single ancestral path of a tree by a directed acyclic graph in which a reticulation may inherit from either of two parents. This flexibility creates an immediate statistical question: even with an arbitrarily long aligned sequence, does the induced distribution of site patterns determine the network topology? If the answer is negative, then no consistent inference procedure can distinguish

the competing histories without additional assumptions or data. The principle that inference should not separate observationally indistinguishable networks has therefore become central in the theory of phylogenetic networks [10].

This paper studies the displayed-tree Jukes–Cantor model on a class of level-2 networks. Level-2 blobs may contain two interacting reticulations, so they are the first class in which reticulation cycles can be genuinely tangled. Englander et al. proved generic identifiability for binary, triangle-free, strongly tree-child, level-2 semi-directed networks under the Jukes–Cantor model [7, Theorem 3.2]. Their triangle-free hypothesis is substantial rather than cosmetic. In a triangle, changing the vertex designated as the reticulation can preserve a full-dimensional region of the observable model. Recent work has clarified both sides of this phenomenon: the three triangle orientations have rich semialgebraic intersections [6], while complete level-1 identifiability can be formulated only modulo triangle redirection [4, Theorem 4.9].

Our purpose is to remove the triangle-free hypothesis for the full binary strongly tree-child level-2 class, while stating exactly what remains unidentifiable. The relevant strength condition must be imposed on the standard semi-directed topology, not merely on one convenient rooted presentation. We call a semi-directed topology *strongly tree-child* when every admissible rooting is tree-child. A preliminary structural theorem shows that any binary level-2 topology admitting even one tree-child rooting already has at most one triangle in each blob. Thus the triangle-count restriction that appeared in an earlier formulation is automatic, rather than a limitation of the identifiability theorem. Generic Jukes–Cantor data identify the bridge tree and every local blob of every binary strongly tree-child level-2 network up to ordinary triangle redirection. The theorem is sharp. If one only asks that at least one rooting be tree-child, an explicit non-triangle ambiguity survives on a full-dimensional regular set for every number of leaves at least four.

1.1 Main contribution

Let $\mathcal{M}_N^{\text{JC}}$ denote the open stochastic model image of a standard semi-directed topology N . We write $N \preceq_{\text{JC}} N'$ when a source-relative open set of full source dimension in $\mathcal{M}_N^{\text{JC}}$ is contained in $\mathcal{M}_{N'}^{\text{JC}}$; this is the one-sided relation relevant to generic topological identifiability. Our principal result is the following.

Theorem (informal form). *Let N and N' be leaf-labelled binary standard semi-directed strongly tree-child level-2 networks. Then $N \preceq_{\text{JC}} N'$ if and only if their labelled homeomorphism-reduced bridge trees agree and each corresponding pair of blobs is port-labelled isomorphic or differs by ordinary triangle redirection. Every surviving one-sided relation is a symmetric full-dimensional regular overlap.*

The proof has six main components.

- (i) A graph-theoretic reduction proves that every weakly tree-child binary level-2 blob has at most one triangle. A blob with two triangles would reduce to the $(1, 2, 2)$ theta, or K_4 minus an edge, and every possible rooting violates tree-childness.
- (ii) A pointwise cut theorem characterizes bridge splits by rank at most four of the appropriate pattern flattening throughout the open Jukes–Cantor parameter domain. The proof includes the delicate case in which the central edge of a crossing quartet joins two active three-port blob tensors.
- (iii) Fourier rank-one factorization across the recovered bridge tree yields projective local tensors. We describe the entire reciprocal gauge group and construct compatible analytic gauge slices. This gives a genuine local product chart rather than only an entrywise gluing identity.

- (iv) Every nonroot cycle or theta blob has a bounded rigid support. One-port restrictions locate extra ports on core segments and two-port restrictions recover their order. Thus arbitrary subdivisions reduce to finite local atlases.
- (v) The last finite case is a seven-outgoing-port cycle-to-theta comparison. It consists of eight free S_4 orbits, or 192 labelled role presentations, with 1,686 standard strongly tree-child completions. The Jukes–Cantor trinet polynomial

$$F_{abc} = r_{ab}r_{ac}r_{bc} - u_{abc}^2$$

was introduced by Englander et al. to distinguish a three-leaf tree from strict level-1 and level-2 trinet [7, Proposition 2.26 in the April 2025 version]; a level- k extension appears in [4, Lemma 5.5]. The polynomial is not new here. What is new is that, after restoring the labels required by a rigid core-3 support, suitable trinet restrictions make F_{abc} a strict separator for every residual seven-port completion.

- (vi) A root-reduction lemma reroots every root-local strongly tree-child factor through an ordinary tree-child path to a pendant edge. Stationarity and reversibility of the Jukes–Cantor model then reduce root factors to the nonroot incoming-port theorem.

The resulting local-to-global theorem excludes coordinated compensation between distant blobs. In an analytic peeling chart, a regular source germ is a Cartesian product of local tensor germs and bridge parameters. Projecting a product box contained in a target model produces a genuine local one-sided containment for each blob. Hence global containment cannot be created by changes that fail locally.

1.2 Relationship to prior work

The exact increment over the closest results is summarized in [table 1](#). The comparison distinguishes algebraic, generic, and pointwise notions of identifiability; they are not interchangeable. In particular, equality of Zariski closures does not imply equality, or even nonempty intersection, of open stochastic images.

Table 1: Closest prior results and the increment proved here.

Reference	Network class	Model	Identifiability notion	Relation to this paper
Holtgreffe et al. [9, Theorem 5 and Corollary 3]	General semi-directed and multi-semi-directed networks	None	Weak/strong tree-child rooting characterizations	Supplies the general rooting framework and omnian characterizations. We prove directly that, at binary level 2, weak tree-childness automatically excludes blobs with two triangles.
Ardiyansyah [1]	Simple and semisimple level-2 networks, including four-leaf types	JC, K2P, K3P	Algebraic variety distinguishability and noncontainment results	Develops Fourier-invariant tools and several distinguishability results for level-2 network varieties. It does not give a global open-stochastic classification of all strongly tree-child level-2 networks.

Reference	Network class	Model	Identifiability notion	Relation to this paper
Englander et al. [7, Theorem 3.2]	Binary triangle-free strongly tree-child level 2	JC	Generic semi-directed topology	Establishes the triangle-free base theorem. The present work covers the entire binary standard strongly tree-child level-2 class and proves that ordinary triangle redirection is the only topology change that can remain compatible on a full-dimensional model region.
Englander et al. [7, Proposition 2.26 in the April 2025 version]	Three-leaf tree versus strict level-1/2 trinet	JC	Pointwise strict inequality	Introduces F_{abc} . We use it on restored rigid-support labels to close the final seven-port level-2 comparison.
Brits et al. [4, Theorem 4.9]	All level-1 networks	JC, K2P, K3P	Full pointwise topology modulo redirected triangles	Gives the complete level-1 result; their level- k trinet inequality [4, Lemma 5.5] generalizes the earlier base inequality. Our result concerns tangled level-2 blobs and is generic, not pointwise.
Currie et al. [6, Theorem 3.1, Corollary 3.3, Lemma 3.4, Theorem 3.5]	Three labelled three-leaf triangle orientations	JC	Complete semialgebraic model geometry	Shows full-dimensional intersections and differences among the three triangle models. We treat their orientation ambiguity as the local move T and prove no further ambiguity in the strong level-2 class.
Cox, Gross, and Martin [5]	Three-sunlet networks	Group-based models	Dimension and partial identifiability	Provides geometric context for triangle-containing group-based network models.
This paper	All binary standard semi-directed S_{TC} level-2 networks	JC	Generic one-sided classification and reconstruction	Recovers the bridge tree and every blob modulo T ; proves the boundary is sharp by an all- n ambiguity in $W_{TC} \setminus S_{TC}$.

Ardiyansyah’s earlier level-2 study developed direct Fourier-invariant comparisons for simple and semisimple level-2 network varieties, including four-leaf cases [1]. It provides important algebraic context but does not subsume the open-stochastic, local-to-global classification proved here.

The semialgebraic description of Currie et al. is especially relevant conceptually. Their Theorem 3.1 gives explicit inequalities for one three-leaf triangle model; Corollary 3.3 translates the pairwise distinguishability conditions to numerical parameters; Lemma 3.4 proves that both an intersection and a model difference contain nonempty open subsets; and Theorem 3.5 concludes that triangle direction is not generically identifiable [6]. Our use of a common regular germ for T is consistent with this geometry, but our global theorem neither asserts nor requires equality of the complete stochastic images of redirected networks.

1.3 Biological interpretation

The theorem describes which aspects of an evolutionary history are, in principle, recoverable from an exact infinite-data site-pattern distribution under the displayed-tree Jukes–Cantor model. In the strong class, the coarse branching architecture given by the bridge tree is recoverable. Within every reticulated component, the cycle/theta generator, the placement of labelled descendant components, and their order along generator sides are recoverable. Ordinary triangle redirection is the only

topology change that can remain observationally compatible on a full-dimensional model region. Thus the generic topology is determined modulo this local move. This is biologically natural: a triangle represents reticulation between very closely related lineages, and its orientation signal can be erased on part of parameter space even while the presence of reticulation remains detectable.

Strong tree-childness is not merely a technical convenience. The sharpness construction exchanges two labelled pendant components in a theta blob. Both resulting standard semi-directed topologies possess tree-child rootings, but not every admissible rooting is tree-child. They are not related by triangle redirection, yet they generate a common full-dimensional regular Jukes–Cantor region. Thus weakening the structural hypothesis crosses an actual identifiability boundary.

1.4 Organization

Section 2 fixes graph and model conventions. The formal theorems are stated in section 3. Sections 4 and 5 develop bridge peeling and pointwise cut recovery. Section 6 proves the automatic triangle bound and bounded-support reconstruction. Section 7 gives the local atlas and seven-port closure. Section 8 proves root reduction, local-to-global completeness, and sufficiency of T. Genericity and reconstruction appear in section 9; the sharpness theorem is proved in section 10. The computer-assisted proof and its independent verification are described in section 11. Exact dimension and sharpness data are collected in the appendices.

2 Networks, reductions, and stochastic models

2.1 Rooted and standard semi-directed networks

Let X be a finite set of taxon labels. A *rooted binary phylogenetic network* on X is a finite directed acyclic graph with no parallel arcs in which: the root has indegree 0 and outdegree 2; each leaf has indegree 1, outdegree 0, and a distinct label in X ; each tree vertex has bidegree $(1, 2)$; and each reticulation has bidegree $(2, 1)$. An edge is called a *reticulation edge* when its head is a reticulation. All other edges are ordinary tree edges.

Definition 2.1 (Standard semi-directed reduction). *Given a rooted binary network, its standard semi-directed reduction is obtained by retaining the arrowheads on reticulation edges, replacing every ordinary edge by an undirected edge, suppressing the root, and then exhaustively suppressing unlabelled ordinary vertices of undirected degree two. If suppression produces the standard parallel root artifact, it is resolved before further degree-two suppression. Isomorphisms preserve taxon labels, undirected edges, and retained arrowheads.*

The reduction forgets ordinary edge directions but does not forget which vertices receive reticulation arrowheads. It is the network analogue of passing from a rooted tree to an unrooted tree under a stationary reversible substitution model. We use N for a standard semi-directed topology and N^+ for one rooted partner.

Definition 2.2 (Admissible rooting). *An admissible rooting of a standard semi-directed topology N is a rooted binary network N^+ whose standard reduction is N . A rooting may insert the root on an undirected edge or, when compatible with the retained arrowhead, on an edge entering a reticulation. All directed edges must form an acyclic orientation with the prescribed binary bidegrees.*

A rooted network is *tree-child* when every internal vertex has a child that is a tree vertex or a leaf. Equivalently in the present binary setting, no internal vertex has two reticulation children and no reticulation has a reticulation child. Three different classes arise.

Definition 2.3 (Three tree-child classes).

$$\begin{aligned} R_{\text{TC}} &= \{N^+ : \text{the supplied rooted DAG } N^+ \text{ is tree-child}\}, \\ W_{\text{TC}} &= \{N : N \text{ has at least one tree-child admissible rooting}\}, \\ S_{\text{TC}} &= \{N : \text{every admissible rooting of } N \text{ is tree-child}\}. \end{aligned}$$

Thus $S_{\text{TC}} \subseteq W_{\text{TC}}$, whereas R_{TC} is a property of a chosen rooted presentation rather than of a semi-directed topology. These notions must not be conflated. This weak/strong rooting distinction agrees with the general semi-directed framework of Holtgrete et al. [9]. The main positive theorem concerns S_{TC} ; the sharpness example lies in $W_{\text{TC}} \setminus S_{\text{TC}}$ and has explicit rooted representatives in R_{TC} .

Lemma 2.4 (Local strong criterion). *For a binary standard semi-directed topology, membership in S_{TC} is equivalent to the following local condition (compare [9, Theorem 5]): every tail of a retained reticulation edge has two incident undirected edges.*

Proof. Suppose a tail v of a retained arrow $v \rightarrow r$ has fewer than two incident undirected edges. Binary degree forces another retained arrow incident at v or forces a rooting in which the only ordinary incident edge points toward v . In the first case v has two reticulation children; in the second, the admissible rooting on the opposite side makes both children reticulations. Thus some admissible rooting is not tree-child.

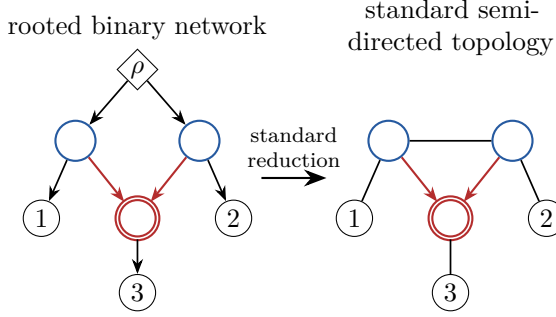
Conversely, assume the local condition. In any admissible rooting, each tail of a retained arrow has at least one ordinary child, because among its two undirected incident edges at most one points toward it. If a reticulation had a reticulation child, its outgoing edge would be a retained arrow with that reticulation as its tail. Its two incoming edges are also retained at the same vertex, so it would have no incident undirected edge, contradicting the local condition. Every internal vertex therefore has a tree-or-leaf child in every admissible rooting. Hence every rooting is tree-child. $\square$

2.2 Blobs, bridges, and generator cores

The underlying undirected multigraph of a standard semi-directed topology is obtained by forgetting retained arrowheads. A *blob* is a maximal biconnected subgraph containing at least one cycle. An edge whose deletion disconnects the topology is a *bridge*. Contracting every blob and suppressing unlabelled degree-two vertices gives the *homeomorphism-reduced bridge tree*. Each component attached to a blob through a bridge is represented locally by a *port*; ports inherit the labels of their descendant leaf blocks.

A network is *level k* when every blob contains at most k reticulations. The main theorem assumes only level at most two; it imposes no separate triangle-count restriction. In [theorem 3.1](#) we prove that every binary level-2 topology in W_{TC} —and hence every topology in S_{TC} —automatically has at most one undirected 3-cycle in each blob.

Suppress within a blob every ordinary tree vertex that carries no port and has degree two in the blob. The resulting directed mixed object is its *ported generator core*. At level two the only nontrivial undirected cores are a cycle and a theta graph: the latter consists of three internally disjoint paths between two branch vertices. Direction and reticulation roles refine the theta graph into four oriented cores. Ordinary port-bearing tree vertices form finite ordered words on the directed core segments. The exact finite list and its proof are given in [section 6](#); [figure 4](#) gives a schematic preview.



Three tree-child notions

R_{TC} : the supplied rooted DAG is tree-child.

W_{TC} : at least one admissible rooting is tree-child.

S_{TC} : every admissible rooting is tree-child.

$S_{TC} \subseteq W_{TC}$, whereas R_{TC} concerns one chosen rooted presentation.

Figure 1: Rooted-to-semi-directed reduction and the three tree-child notions. Ordinary edges lose their directions, whereas arrowheads entering reticulations are retained. Membership in S_{TC} is a property of the standard semi-directed topology, not of one preferred rooting.

Definition 2.5 (Ordinary triangle redirection). *Let N contain a triangle C in a blob. For the level-2 topologies in W_{TC} considered below, C is necessarily the unique triangle in that blob by theorem 3.1. An ordinary triangle redirection T changes which vertex of C receives the two retained arrowheads, while preserving the labelled underlying graph and every arrowhead outside C . The operation is permitted only when both endpoints are valid standard semi-directed topologies in the class under consideration.*

This is a move between semi-directed topologies, not a change of an unobservable root position. The three external arms of a valid S_{TC} triangle are ordinary undirected edges; this fact is proved in theorem 8.5.

2.3 The Jukes–Cantor network model

Identify the four nucleotide states with the group

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The root distribution is uniform. Each network edge e carries one nontrivial Fourier multiplier $x_e \in (0, 1)$. In probability coordinates the transition matrix is

$$T_e(a, b) = \begin{cases} (1 + 3x_e)/4, & a = b, \\ (1 - x_e)/4, & a \neq b. \end{cases}$$

Equivalently, the off-diagonal substitution probability is $q_e = (1 - x_e)/4 \in (0, 1/4)$. Every reticulation r has an inheritance probability $\lambda_r \in (0, 1)$. Independently at each reticulation, one incoming parent is selected; after deleting the unselected incoming edges, suppressing degree-two artifacts gives a displayed tree. The network distribution is the inheritance-weighted mixture of its displayed-tree distributions.

The *open parameter space* is

$$\Theta_0(N) = (0, 1)^{E(N)} \times (0, 1)^{R(N)}.$$

No zero multiplier, zero-length edge, infinitely long edge, or degenerate inheritance probability is allowed.

For a zero-sum character assignment $g = (g_i)_{i \in X} \in G^X$ with $\bigoplus_i g_i = 0$, let A_e be either side of the split induced by edge e in a displayed tree. The Fourier coordinate is

$$\hat{p}_N(g) = \sum_{\sigma} w_{\sigma} \prod_{e \in T_{\sigma}} x_e^{\mathbf{1}_{[\bigoplus_{i \in A_e} g_i \neq 0]}}, \quad (1)$$

where σ ranges over parent selections and w_{σ} is the corresponding product of inheritance probabilities and complements. Coordinates with nonzero total character sum vanish. Formula (1) follows from the standard group-based Fourier transform [8, 11, 12].

The Fourier transform is invertible on the ambient coordinate space. Hence equality, rank, dimension, and semialgebraic questions may be studied in either Fourier or pattern-probability coordinates. We use normalized tensors with $\hat{p}(0, \dots, 0) = 1$.

2.4 Model images and observational relations

Let

$$\phi_N : \Theta_0(N) \longrightarrow \mathbb{R}^{4^{|X|}}$$

be the site-pattern parameterization, and let

$$\mathcal{M}_N^{\text{JC}} = \phi_N(\Theta_0(N))$$

be its open stochastic image. Its complex Zariski closure is denoted

$$\mathcal{V}_N^{\text{JC}} = \overline{\mathcal{M}_N^{\text{JC}}}^Z \subseteq \mathbb{C}^{4^{|X|}}.$$

The closure is irreducible because it is the closure of the polynomial image of an irreducible affine parameter space. We nevertheless keep equality of closures separate from equality or intersection of stochastic images.

A parameter point is *regular* when the Jacobian rank of ϕ_N is locally maximal. The model dimension is the common maximal real/complex Jacobian rank,

$$\dim \mathcal{M}_N^{\text{JC}} = \dim_{\mathbb{C}} \mathcal{V}_N^{\text{JC}}.$$

The equality holds because the real open cube is Zariski dense and the parameterization has real coefficients.

Definition 2.6 (Symmetric overlap and one-sided containment). *Write $N \bowtie_{\text{JC}} N'$ if there is a distribution $p \in \mathcal{M}_N^{\text{JC}} \cap \mathcal{M}_{N'}^{\text{JC}}$ that is regular for both models, the local dimensions agree, and the*

intersection contains a relative neighborhood of p in both images. Write

$$N \preceq_{\text{JC}} N'$$

if $\mathcal{M}_N^{\text{JC}} \cap \mathcal{M}_{N'}^{\text{JC}}$ contains a semialgebraic subset of full dimension $\dim \mathcal{M}_N^{\text{JC}}$, equivalently a nonempty relatively open regular subset of the source image by [theorem 9.1](#) below. The target may have larger dimension and a target preimage need not be regular.

This orientation is important. If $\mathcal{V}_N \subseteq \mathcal{V}_{N'}$, then

$$I(\mathcal{V}_{N'}) \subseteq I(\mathcal{V}_N).$$

A polynomial that vanishes on the source and is strictly nonzero throughout the target proves disjointness of the open stochastic images, even though it is not a target-ideal witness for algebraic noncontainment.

Definition 2.7 (Generic topological identifiability). *Fix a finite topology N . The topology is generically identifiable modulo $\mathbb{T}$ if there is a proper algebraic subset $Z \subsetneq \Theta_0(N)$ such that, for every $\theta \notin Z$, equality*

$$\phi_N(\theta) = \phi_{N'}(\theta')$$

with another valid topology N' implies that N' is labelled-isomorphic to a topology obtained from N by valid ordinary triangle redirections.

The phrase *generic* always means outside a proper algebraic parameter locus. It does not mean merely that one favorable Euclidean-open set exists.

3 Main results

We first record the structural fact that removes the triangle-count hypothesis from the statistical theorem.

Theorem 3.1 (Automatic triangle bound). *Every binary standard semi-directed level-2 topology in W_{TC} has at most one triangle in each blob. In particular, the conclusion holds for every topology in S_{TC} .*

The proof is purely graph theoretic and appears in [section 6.2](#). It reduces a hypothetical two-triangle blob to the $(1, 2, 2)$ theta core and excludes every possible tree-child rooting.

We now state the observational classification. All networks in [theorems 3.2](#) and [3.3](#) are binary standard semi-directed S_{TC} level-2 topologies on the same fixed leaf set.

Theorem 3.2 (One-sided observational classification). *Let N and N' be networks in the stated class. Then*

$$N \preceq_{\text{JC}} N'$$

if and only if the following conditions hold.

- (a) *Their labelled homeomorphism-reduced bridge trees are isomorphic.*
- (b) *Under the induced correspondence of ports, every pair of corresponding local factors is port-labelled isomorphic or related by ordinary triangle redirection $\mathbb{T}$.*

Every surviving one-sided relation is a symmetric full-dimensional regular overlap. In particular, there are no proper one-sided generic containments between distinct local generator types or port placements.

The necessity is proved in [section 8](#). The pointwise cut theorem forces the same bridge tree; analytic peeling localizes containment to each blob; the local atlas then leaves only isomorphism and T . Sufficiency follows from the common regular triangle tensor germ and positive contextual gluing.

Corollary 3.3 (Generic identifiability modulo triangles). *For every fixed topology N in the stated class, there is a proper algebraic subset $Z_N \subsetneq \Theta_0(N)$ such that the exact JC distribution at every $\theta \in \Theta_0(N) \setminus Z_N$ determines the standard semi-directed topology up to labelled isomorphism and valid ordinary triangle redirection.*

Corollary 3.4 (Recoverable structure). *At a generic parameter point the exact JC distribution determines:*

- (i) *the labelled homeomorphism-reduced bridge tree;*
- (ii) *whether each blob is a cycle or a theta blob;*
- (iii) *the oriented theta core modulo the local triangle move;*
- (iv) *the labelled descendant block attached at every port;*
- (v) *the directed core segment containing every port and the total order of ports along each segment.*

Ordinary triangle redirection is the only topology change that can remain observationally compatible on a full-dimensional model region. Equivalently, the generic topology is determined modulo ordinary triangle redirection.

The conclusion concerns semi-directed topology. Rooted presentations remain nonidentifiable under the stationary reversible JC model. Moreover, T -related endpoints are proved to share a full-dimensional regular stochastic germ, not necessarily the same complete open stochastic image.

The positive theorem is sharp with respect to the tree-child convention.

Theorem 3.5 (Sharpness of strong tree-childness). *For every $n \geq 4$ there are two nonisomorphic, non- T -equivalent binary level-2 standard semi-directed networks N_n, N'_n such that*

$$N_n, N'_n \in W_{\mathsf{TC}} \setminus S_{\mathsf{TC}}, \quad N_n \bowtie_{\mathsf{JC}} N'_n.$$

The common local model dimension is $2n$. Both topologies have rooted tree-child representatives, but each has admissible rootings that are not tree-child.

The explicit four-leaf pair and its all- n leaf-substitution extension are given in [section 10](#). Together with [theorem 3.1](#), this establishes the sharp boundary statement

S_{TC} is generically JC-identifiable modulo T ,	W_{TC} is not.
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for all binary standard semi-directed level-2 networks.

Remark 3.6 (Scope). *The theorem does not address level greater than two, substitution models richer than JC, a positive classification of the full weakly tree-child class, or equality of complete stochastic images under T . Blobs with multiple triangles are not an omitted level-2 case: [theorem 3.1](#) proves that no binary topology in W_{TC} can contain one.*

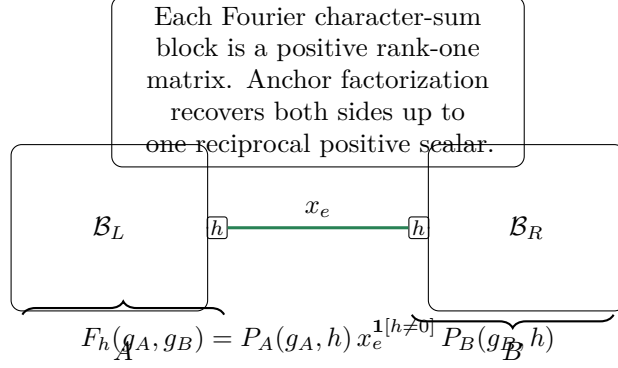


Figure 2: Bridge factorization in Fourier coordinates. Peeling the bridge tree gives projective local tensors and a reciprocal gauge for each bridge; because the bridge graph is a tree, no gauge can circulate around a cycle of components.

4 Fourier factorization and analytic bridge peeling

This section develops the local coordinates used in the global proof. The key point is not merely that a bridge tensor factors entrywise. We identify the complete reciprocal gauge group, choose analytic gauge slices, and prove a local product theorem in which local tensor variations and bridge variations are independent at a regular point.

4.1 Rank-one character blocks

Let e be a bridge of a network N , inducing the nontrivial leaf split $A \mid B$. Cut e at an interior point and regard the two halves as boundary ports. For a zero-sum assignment $g = (g_A, g_B)$ put

$$h = \bigoplus_{i \in A} g_i = \bigoplus_{j \in B} g_j.$$

The equality follows because the group has exponent two and the total sum is zero.

Proposition 4.1 (Bridge factorization). *At every open JC parameter point, the Fourier tensor satisfies*

$$F_h(g_A, g_B) = P_A(g_A, h) x_e^{1[h \neq 0]} P_B(g_B, h), \quad (2)$$

where P_A and P_B are positive boundary tensors of the two cut components. For fixed h , the matrix with rows indexed by g_A and columns indexed by g_B has rank one and strictly positive entries.

Proof. Condition on the state at an interior point of e . In probability coordinates the two sides are conditionally independent given this four-state variable. Applying the group Fourier transform diagonalizes the transition kernel on e and separates the character sum h . The trivial character has edge multiplier one; every nontrivial character has multiplier x_e . This gives (2). Positivity follows from $x_e \in (0, 1)$ and positivity of every stochastic transition and inheritance weight. $\square$

4.2 The reciprocal gauge group

For a fixed character block, a positive rank-one matrix $F_h = L_h R_h^\top$ determines its two factors only up to

$$(L_h, R_h) \mapsto (c_h L_h, c_h^{-1} R_h), \quad c_h > 0.$$

Normalization fixes $c_0 = 1$. The JC symmetry identifies the remaining three scalars.

Lemma 4.2 (Complete bridge gauge). *The complete factorization ambiguity in [theorem 4.1](#) is one positive scalar c_e per bridge:*

$$P_A(g_A, h) \mapsto c_e^{1[h \neq 0]} P_A(g_A, h), \quad P_B(g_B, h) \mapsto c_e^{-1[h \neq 0]} P_B(g_B, h). \quad (3)$$

There are no additional character-dependent gauges compatible with the JC symmetry.

Proof. Rank-one uniqueness gives a scalar c_h for each character-sum block. The normalized zero block has $c_0 = 1$. Every local JC tensor is invariant under the action of $\text{Aut}(G) \cong S_3$ on the nonzero characters. Applying an automorphism to the factorization of a positive entry shows $c_{\alpha h} = c_h$. Since the action is transitive on $G \setminus \{0\}$, the three nonzero scalars coincide. This common value is c_e . $\square$

Because the bridge graph is a tree, the gauges in (3) cannot circulate. Choose a root component of the bridge tree and peel outward. At each cut, fix one positive anchor entry on the parent side and absorb the reciprocal scalar into the child side. The process uniquely fixes every local projective tensor.

Lemma 4.3 (Anchor reconstruction). *Let $F = (F_{ij})$ be a positive rank-one matrix and fix an anchor (i_0, j_0) . Then*

$$L_i = F_{ij_0}, \quad R_j = \frac{F_{i_0 j}}{F_{i_0 j_0}}$$

reconstruct a factorization $F_{ij} = L_i R_j$. These formulas are real analytic on the positive rank-one locus.

Proof. The rank-one identities give $F_{ij} F_{i_0 j_0} = F_{ij_0} F_{i_0 j}$. Positivity makes the denominator nonzero. Analyticity is immediate. $\square$

4.3 Analytic peeling charts

Let $\mathcal{T}$ be a fixed labelled bridge tree. For each vertex v of $\mathcal{T}$, let $\mathcal{L}_v$ be the local boundary-tensor model of its component, with one boundary index for each incident bridge. Let J index the bridges. The contraction map

$$\Gamma : \left(\prod_v \mathcal{L}_v \right) \times (0, 1)^J \longrightarrow \mathbb{R}^{4^{|X|}}$$

contracts matching boundary characters and inserts the bridge multipliers.

The local models contain redundant arm scalings. Fix positive anchor entries for each incident bridge and impose one analytic normalization equation for each reciprocal gauge. Denote the resulting slices by $\mathcal{S}_v$.

Theorem 4.4 (Analytic bridge-peeling chart). *Let p be a positive regular point of a network model with bridge tree $\mathcal{T}$. After choosing positive anchors, there are neighborhoods $U_v \subseteq \mathcal{S}_v$ of the extracted*

local tensors and intervals $I_e \subset (0, 1)$ such that the contraction map restricts to a real-analytic diffeomorphism from

$$\prod_{v \in V(\mathcal{T})} U_v \times \prod_{e \in E(\mathcal{T})} I_e \quad (4)$$

onto a relative neighborhood of p in the model image. The only quotient removed in forming the slices is the reciprocal gauge group of [theorem 4.2](#).

Proof. Peel the bridge tree from its leaves. At a leaf component, every character block of the incident cut flattening is a positive rank-one matrix. The anchor formulas of [theorem 4.3](#) recover the boundary tensor of the leaf component and the complementary tensor of the remaining network, up to the single JC gauge. The chosen normalization fixes this gauge analytically. Remove the leaf component and iterate. Since the bridge graph is a tree, every bridge is processed exactly once and no consistency equation appears around a cycle.

The reconstruction formulas define a real-analytic left inverse to Γ on the positive rank-one locus satisfying the fixed anchors. Conversely, contraction gives a right inverse. At a regular point the source tangent rank equals the sum of the sliced local ranks and the bridge directions: the Jacobian is block triangular after ordering variables by the peeling sequence, and each diagonal block is the corresponding local tangent block or a nonzero bridge derivative. The constant-rank theorem therefore gives the claimed product neighborhood and analytic diffeomorphism. $\square$

The point of the last statement is that the product structure is intrinsic near a regular source point. We do not assume that a target preimage of the same distribution is regular.

4.4 Tripod gluing and contextual contraction

A particularly useful special case glues two boundary tensors through a new tripod. Let P be a normalized positive Fourier tensor on $X \cup \{a\}$ and Q one on $Y \cup \{b\}$. Delete the terminal leaves a, b , join the exposed edges to a new tree vertex, and attach a new leaf c with multiplier $z \in (0, 1)$. With

$$h_X = \bigoplus_{i \in X} g_i, \quad h_Y = \bigoplus_{j \in Y} g_j, \quad h_X \oplus h_Y \oplus g_c = 0,$$

the glued tensor is

$$\hat{\Gamma}(g_X, g_Y, g_c) = P(g_X, h_X)Q(g_Y, h_Y)z^{1[g_c \neq 0]}. \quad (5)$$

For a nonzero character h , choose assignments with $h_X = h_Y = h$ and define

$$U = \hat{\Gamma}(g_X, 0, h), \quad V = \hat{\Gamma}(0, g_Y, h), \quad W = \hat{\Gamma}(g_X, g_Y, 0).$$

Then

$$z = \sqrt{\frac{UV}{W}}, \quad P(g_X, h) = \frac{U}{z}, \quad Q(g_Y, h) = \frac{V}{z}. \quad (6)$$

The positive square root is unique. These identities prove positivity, dimension additivity, and propagation of a local regular overlap through identical attached components. More generally, any unchanged network context defines an analytic multilinear contraction in the boundary tensor. This observation will be used for an embedded triangle whose context reconnects two of its three arms.

5 Pointwise recovery of bridge splits

Generic split recovery from flattening ranks is standard in latent-tree models, but one-sided containment requires a stronger statement. A source model could, a priori, lie inside the rank-exceptional locus of a larger target. We therefore prove a pointwise theorem on the complete open JC domain.

For a leaf split $A \mid B$, write $\text{Flat}_{A|B}(p)$ for the ordinary pattern-probability flattening. The Fourier transform acts invertibly and separately on the two sides, so its rank equals the rank of the corresponding Fourier flattening.

Theorem 5.1 (Pointwise cut characterization). *Let N be a binary standard semi-directed S_{TC} level-2 network, and let $p \in \mathcal{M}_N^{\text{JC}}$. For every nontrivial leaf split $A \mid B$,*

$$\text{rank Flat}_{A|B}(p) \leq 4 \iff A \mid B \text{ is induced by a bridge of } N.$$

This holds at every point of the open parameter domain, not only generically.

The forward implication for a true bridge is immediate from conditional independence at the hidden four-state variable. The converse is local. Choose a crossing quartet and contract irrelevant pendant paths into effective multipliers. If the crossing is contained in one blob, a finite one-active atlas supplies a nonzero wrong-split minor unless the tested split is a true bridge in every switching. If the crossing passes through a bridge joining two active blob endpoints, the following endpoint calculation applies.

5.1 Three-boundary endpoint tensors

Normalize a three-boundary JC tensor by its zero coordinate and choose a fixed nonzero character. By JC symmetry, the four relevant coordinates may be written

$$(1, a, b, c, t),$$

where a, b, c are the three pair coordinates and t is the all-distinct zero-sum coordinate. Put

$$\Delta = abc - t^2, \quad \Gamma = a - bc.$$

The roles of a, b, c can be permuted according to which boundary is designated as the bridge interface.

Lemma 5.2 (Endpoint dichotomy). *Every open three-boundary endpoint tensor arising from an admissible root, cycle, or theta completion in the stated class satisfies exactly one of*

$$\Delta > 0, \tag{7}$$

$$\Delta = 0 \quad \text{and} \quad \Gamma > 0. \tag{8}$$

The complete canonical endpoint universe has 177 types: 151 satisfy (7) and 26 satisfy (8).

Proof. Suppress ordinary boundary paths into products of edge multipliers. By the level-2 generator theorem and the strong local criterion, every possible endpoint is obtained from one cycle or theta core, one admissible interface role, and a minimal strong repair. The independent enumerators described in [section 11](#) canonicalize these objects before evaluating their displayed-tree tensors.

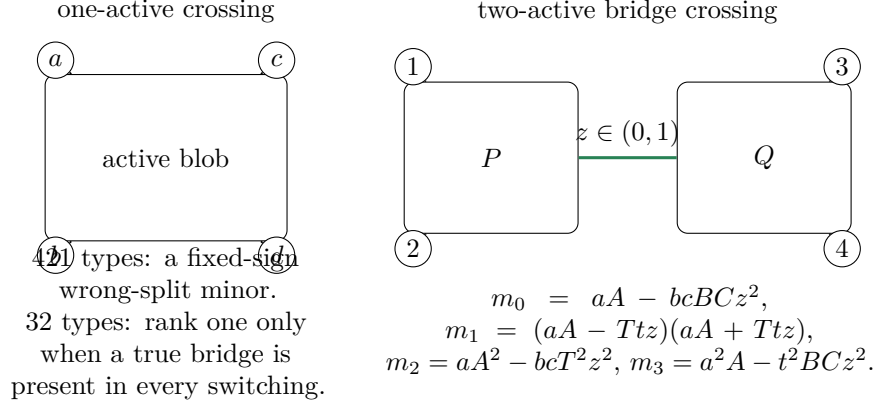


Figure 3: The two configurations in the pointwise cut theorem. The second is the previously delicate case: the crossing quartet’s central bridge joins two active local tensors.

For each type, Δ and, when necessary, Γ are expanded in the tensor-product Bernstein basis on the edge and inheritance cube. In 151 cases every Bernstein coefficient of Δ is nonnegative and at least one coefficient active in the open cube is positive. In the remaining 26 cases Δ is the zero polynomial and the same test applied to Γ is strictly positive. Since every Bernstein basis function is positive in the open cube, the stated signs follow. The reviewer independently regenerates the canonical types, rather than reading the primary list as input. $\square$

The identity $\Delta = 0$ is the tree-like boundary equation. The strict inequality $a > bc$ in the second branch is the extra information needed when two such tree-like endpoints are joined.

5.2 One active crossing

Suppose a crossing quartet meets one active blob and otherwise passes through ordinary bridge paths. After path contraction, there are four boundary ports and a proposed wrong split.

Lemma 5.3 (One-active crossing). *Among the 453 canonical one-active crossing tensor types, 421 possess a Fourier minor of the wrong-split flattening with a fixed nonzero sign throughout the open parameter cube. In the remaining 32 types every wrong-split block is rank one identically, and the tested split is a genuine bridge split in every displayed switching.*

Proof. The universe is generated from the primitive cycle/four-theta cores, all admissible interface roles, and every minimal strong repair. Canonical mixed-graph encodings remove duplicates. For each non-bridge type the verifier enumerates the 2×2 Fourier block minors and selects one whose polynomial is nonzero. Exact factorization or tensor-product Bernstein expansion proves a fixed sign over the open cube. The 32 residual signatures are checked graphically: deleting the designated edge disconnects the selected leaf pairs in every parent switching, so the rank-one behavior comes from a true bridge rather than an accidental degeneracy. $\square$

5.3 Two active endpoints

Now suppose the crossing quartet has two active endpoint tensors joined by a bridge of multiplier $z \in (0, 1)$. Let their normalized coordinates be

$$(a, b, c, t), \quad (A, B, C, T).$$

Direct Fourier contraction gives four wrong-split 2×2 minors:

$$m_0 = aA - bcBCz^2, \tag{9}$$

$$m_1 = (aA - Ttz)(aA + Ttz), \tag{10}$$

$$m_2 = aA^2 - bcT^2z^2, \tag{11}$$

$$m_3 = a^2A - t^2BCz^2. \tag{12}$$

All factors not shown have been divided out only after being verified strictly positive on the open domain.

Lemma 5.4 (Two-active contradiction). *The four minors (9)–(12) cannot vanish simultaneously for a wrong split.*

Proof. Assume all four vanish. Positivity makes $aA + Ttz > 0$, so (10) gives

$$aA = Ttz.$$

Combining this identity with (11) and (12), and cancelling positive factors, yields

$$ABC = T^2, \quad abc = t^2.$$

Thus both endpoint tensors lie in the zero branch of [theorem 5.2](#), and therefore

$$a > bc, \quad A > BC.$$

Multiplying gives $aA > bcBC$. On the other hand, (9) gives

$$aA = bcBCz^2 < bcBC,$$

because $0 < z < 1$. This is impossible. □

Proof of theorem 5.1. If $A \mid B$ is a bridge split, [theorem 4.1](#) gives rank at most four. Conversely, take a crossing quartet for a non-bridge split in the homeomorphism-reduced bridge tree. Contract ordinary paths and suppress irrelevant components. The central crossing either lies in one active factor, where [theorem 5.3](#) gives a nonzero wrong-split minor, or joins two active endpoints across a bridge, where [theorem 5.4](#) gives a contradiction to wrong-split rank one. The quartet minor is a minor of a marginal of the full flattening; rank greater than four for the quartet therefore forces rank greater than four globally. □

Corollary 5.5 (Cut preservation under one-sided containment). *If $N \preceq_{\text{JC}} N'$, then N and N' have the same nontrivial bridge splits and hence isomorphic labelled homeomorphism-reduced bridge trees.*

Proof. Choose a source-relative open subset contained in the target. At every common open stochastic point, [theorem 5.1](#) characterizes the cut splits from the distribution alone. Thus the split

sets agree. A compatible set of labelled nontrivial splits determines the unique homeomorphism-reduced tree: after fixing a reference leaf, the sides not containing it form a laminar family ordered by inclusion. $\square$

6 Generator cores and bounded-support reconstruction

The local topology space is infinite because any directed generator segment may carry an arbitrarily long ordered word of port-bearing tree vertices. The finite computation becomes complete only after a structural theorem shows that bounded restrictions recover every such word. We derive that theorem here from the level-2 and strong tree-child conditions.

6.1 The finite core list

For a connected blob B , its cyclomatic number is

$$\beta(B) = |E(B)| - |V(B)| + 1.$$

Choose any rooted partner and delete one incoming edge at each reticulation, as in the construction of a displayed tree. The restriction of the resulting undirected graph to B is a forest. If $r(B)$ is the number of reticulations in B , at most $r(B)$ edges of B were deleted, whereas at least $\beta(B)$ edge deletions are required to turn a connected graph into a forest. Hence

$$\beta(B) \leq r(B). \tag{13}$$

A level-2 blob therefore has cyclomatic number at most two. Delete pendant cut-edge components but mark their attachment sites as ports, and suppress every unported ordinary vertex of degree two in the blob.

Proposition 6.1 (Level-2 core theorem). *Every nontrivial reduced level-2 blob core is one of the following.*

- (i) *A cycle core with one source event and one path-sink reticulation event.*
- (ii) *A theta core: two branch vertices joined by three internally disjoint paths, with one source event and two reticulation events. Up to branch reversal and path permutation, acyclicity, reachability, and binary degree constraints give exactly four directed theta cores.*

Every full ported blob is obtained uniquely from one such directed core by inserting ordered words of ordinary port-bearing tree vertices on its directed segments and adding the compulsory children of path-sink reticulations.

Proof. Equation (13) gives $\beta(B) \leq 2$. If $\beta(B) = 1$, biconnectivity forces B to reduce to a cycle. Suppose $\beta(B) = 2$. After suppressing degree-two path vertices, every remaining core vertex has degree three: biconnectivity gives minimum degree at least two, all degree-two vertices have been suppressed, and binary network degrees give maximum degree three. Thus the kernel is cubic. Writing v and e for its numbers of vertices and edges,

$$e - v + 1 = 2, \quad 2e = 3v,$$

so $v = 2$ and $e = 3$. The three kernel edges expand to three internally disjoint paths between two poles, which is a theta graph.

For orientations, record along the three theta paths the source event S and the one or two path-sink events X , allowing a reticulation to occupy a branch vertex. Binary indegree/outdegree constraints fix each segment direction adjacent to an event; empty paths have two possible directions. Reject directed cycles and orientations not reachable from S . Exhaustive branch reversal and path permutation reduce the 24 valid raw orientations to four core types. This elementary enumeration is regenerated by `core_enumerator.py`. Re-inserting suppressed degree-two vertices produces ordered words on directed segments, and each such vertex carries exactly one bridge port in a binary network. $\square$

6.2 Automatic exclusion of multi-triangle blobs

The finite core list alone does not assume a bound on the number of triangles. The next result shows that weak tree-child rootability supplies that bound automatically.

Lemma 6.2 (A bridge cannot enter a reticulation). *Let D be a rooted acyclic phylogenetic network. If an arc $x \rightarrow r$ enters a reticulation r , then its underlying undirected edge is not a bridge.*

Proof. Let $y \rightarrow r$ be the other incoming arc. If the undirected edge xr were a bridge, removing it would place the root and x in one component and r, y in the other. A directed root-to- y path must cross the bridge through $x \rightarrow r$ and then continue from r to y . Together with $y \rightarrow r$, this creates a directed cycle, a contradiction. $\square$

Proof of theorem 3.1. Let $N \in W_{TC}$ and choose a tree-child admissible rooting D . By theorem 6.1, every nontrivial level-2 blob reduces to a cycle or theta core. A cycle core has only one simple cycle, so consider a theta blob. Let the three internally disjoint pole-to-pole paths have lengths

$$1 \leq \ell_1 \leq \ell_2 \leq \ell_3.$$

Its three simple cycles have lengths $\ell_i + \ell_j$. Because the underlying binary network is simple, at most one theta path has length one. If two distinct cycles are triangles, two pairwise sums equal three. The only possibility is

$$(\ell_1, \ell_2, \ell_3) = (1, 2, 2).$$

Thus the blob is the graph K_4 minus one edge. Write u, v for its poles, a, b for the internal vertices of the two length-two paths, and

$$E(B) = \{uv, ua, av, ub, bv\}.$$

The vertices a, b each have one external bridge, while all three edges incident with either pole lie in the blob.

Here $\beta(B) = 2$, so (13) and the level-2 bound force exactly two reticulations in B . If the two reticulations are adjacent, then either their common edge is directed from one reticulation to the other, giving a reticulation child, or the root subdivides that edge, giving the root two reticulation children. Both alternatives violate tree-childness.

The only nonadjacent vertex pair is $\{a, b\}$. Suppose therefore that a and b are the reticulations. By theorem 6.2, their external bridges are their outgoing edges; neither bridge can bring the root into the blob. The binary root must consequently subdivide one of the five internal edges. If it subdivides uv , then both poles have children a and b . If it subdivides an edge between a pole and one reticulation, the opposite pole is a parent of both a and b . In every case a tree vertex has two

reticulation children, again contradicting tree-childness. Hence no tree-child rooting exists, contrary to $N \in W_{TC}$. $\square$

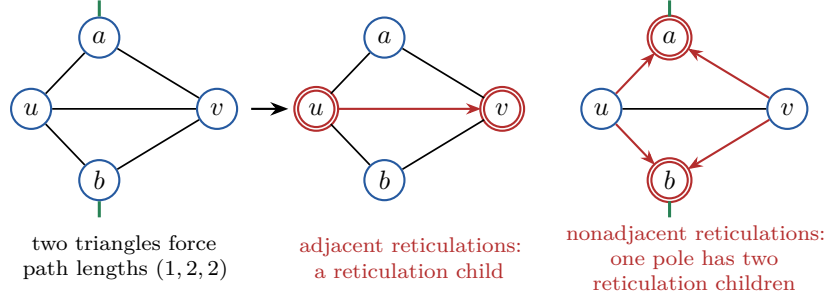


Figure 5: Automatic exclusion of multiple triangles. A simple theta with two triangle cycles must have path lengths (1, 2, 2) and underlying graph K_4 minus the edge ab . Adjacent reticulations force a reticulation child (or a root with two reticulation children). The only nonadjacent pair is a, b ; their external bridges must point away from them, and every internal root position leaves one pole with both reticulations as children.

Remark 6.3 (Independent finite replay). *The structural proof is independently checked by a primitive rooting census that imports none of the atlas code. Across all external and internal root sites on K_4 minus an edge, it finds 25 binary acyclic rooted presentations in seven graph-symmetry orbits and zero tree-child presentations. A separately written C++ implementation obtains the same zero-survivor conclusion. These computations audit the case split; they are not a substitute for the proof above.*

6.3 Strong repairs and rigid supports

A directed core may fail the tree-child condition before ports are inserted. An inserted ordinary port vertex on a segment repairs its tail by providing a tree child. For each core, a *minimum strong repair* is a minimum set of segments whose subdivision makes the local rooted factor tree-child and prevents a reticulation child of a reticulation.

For the four theta cores, the minimum rigid-support sizes are

$$3, \quad 4, \quad 3, \quad 3, \tag{14}$$

where every path-sink child is included and one ordinary port is chosen on each segment of a contained minimum repair. The cycle core has a two-port support: its sink child and one ordinary side anchor.

Lemma 6.4 (Monotonicity of strong occupancy). *If a set of occupied directed segments contains a minimum strong repair, then inserting additional port vertices on any segments preserves the local tree-child conditions. Conversely, every full strongly tree-child expansion contains at least one minimum repair.*

Proof. Subdividing a segment by an ordinary tree vertex changes no reticulation indegree and introduces a vertex with one continuation child and one port child. It therefore cannot destroy a tree child already present. The obligations in the unexpanded core are finite: every source/tree vertex lacking an ordinary child requires at least one of its outgoing segments to be occupied, and

every reticulation whose core child is reticulate requires its unique outgoing segment to be occupied. A full strong expansion satisfies all obligations; deleting occupied segments greedily while retaining satisfaction ends at a minimal repair. $\square$

Definition 6.5 (Rigid support). *A rigid support of a full ported blob consists of all compulsory path-sink children together with one labelled port on each segment of one contained minimum strong repair. We retain the labels of the corresponding descendant bridge components.*

The exact directed roles of a rigid support have trivial pointwise stabilizer except for the simultaneous side swap of the cycle core. This prevents independent local automorphisms from making probe identifications inconsistent.

6.4 One- and two-port probes

Let S be a fixed labelled rigid support. If p is an additional port, restrict the blob to $S \cup \{p\}$ and suppress unlabelled degree-two vertices. The resulting support-plus-one topology records the directed segment containing p . If p, q lie on the same segment, the support-plus-two restriction to $S \cup \{p, q\}$ records their order; if they lie on different segments, their segment locations already distinguish them.

Theorem 6.6 (Bounded-support reconstruction). *Every finite port-labelled strongly tree-child nonroot cycle or theta blob is determined, up to port-labelled core isomorphism and ordinary triangle redirection, by its restrictions to:*

- (i) *one rigid support;*
- (ii) *that support plus each single additional port;*
- (iii) *that support plus each pair of additional ports.*

Consequently, theta blobs require at most six outgoing ports and cycle blobs at most four outgoing ports for structural reconstruction. Including the distinguished incoming boundary gives at most seven and five boundary ports, respectively.

Proof. By [theorem 6.4](#), every full blob contains a rigid support of the stated size. Its labelled roles identify the directed core and one anchor on every required segment. A support-plus-one restriction determines on which core segment each extra label lies: after suppression, the label appears in the unique interval between the same pair of core/support reference points. For two labels on one segment, the support-plus-two restriction distinguishes the two possible directed orders. These pairwise comparisons form a transitive tournament because they arise from an actual directed word, and hence reconstruct the unique total order on each segment. The cycle's only core automorphism swaps its two sides simultaneously; labelled anchors either break this symmetry or leave exactly that allowed swap. Theta automorphisms are accounted for in the canonical core encoding and, pointwise on a labelled support, are trivial. $\square$

6.5 Weak restrictions and completion

A support selected in a source blob need not remain strong when the same descendant blocks are marginalized in a competing target. Omitted path-sink children and omitted repair ports become unobserved dummy components. Completeness therefore requires a weak-target completion theorem.

Lemma 6.7 (Completion of selected restrictions). *Every selected restriction induced from a full strongly tree-child cycle or theta blob is obtained by:*

- (a) *assigning selected labels to arbitrary directed core segments and an arbitrary subset of path-sink children;*
 - (b) *adding an unobserved child at every omitted path sink; and*
 - (c) *adding unobserved port vertices on a minimum set of segments that repairs the full topology.*
- After marginalization, different choices of dummy vertices on the same segment affect the JC tensor only through products of their edge multipliers.*

Proof. Suppress all unselected port-bearing vertices in the full target. What remains is the directed core, the selected labels, and possibly gaps at compulsory sink children or tree-child obligations. Restoring one dummy leaf at every omitted sink and one ordinary dummy port on each missing repair segment produces a full strong extension. Conversely, deleting the dummy leaves recovers the selected restriction. In the Fourier parameterization, consecutive ordinary edges with identical displayed-tree descendant-mask vectors occur in every monomial together, so their parameters enter only through their product. The product map $(0, 1)^k \rightarrow (0, 1)$ is surjective. $\square$

For most selected restrictions, a completion needs at most two new outgoing labels. Core 3 has a genuine three-label case. The completion census is

additional rigid-support labels	labelled presentations
1	2,256
2	1,920
3	192

Thus seven outgoing ports are both sufficient by [theorem 6.6](#) and necessary for the finite weak-target classification. The 192 residual records are classified in [section 7.3](#).

7 The complete nonroot local atlas

We now classify one-sided containment among arbitrary nonroot local factors. The proof is computer-assisted only in its finite portions: generation and canonicalization of bounded supports, exact Fourier expansion, zero/nonzero invariant signatures, strict positivity, and Jacobian minors. The promotion to arbitrary subdivisions is the structural theorem of [theorem 6.6](#).

7.1 Finite atlas protocol

Every bounded record consists of:

- (i) a primitive cycle or oriented theta core;
- (ii) a selected subset of path-sink ports;
- (iii) a weak composition recording the number of selected ordinary ports on each directed core segment;
- (iv) a minimum dummy repair completing the target to a full strongly tree-child factor; and
- (v) a relative labelling of the observed outgoing ports and the distinguished incoming boundary.

Canonical mixed-graph codes preserve leaf labels, undirected edges, and retained arrowheads, and quotient ordinary triangle direction exactly when forming a T class. The generator proves completeness by iterating over the finite lists in [theorems 6.1](#) and [6.7](#); duplicate-freeness is checked by requiring one record per canonical code.

For each record, the displayed-tree Fourier parameterization is expanded from (1). On a four-boundary marginal, an edge is represented by its descendant mask in each parent-choice switching.

Table 2: Canonical bounded local atlases. “Weak” includes selected restrictions completed by unobserved sink children and repair ports.

family	outgoing ports	strong classes	weak classes	strict directed pairs
theta	5	8,520	16,590	18,480
theta, support-size-four core	6	10,980	218,925	21,960
cycle	3	9	12	0
cycle	4	48	63	0
cycle	5	—	390	0
cycle	6	—	2,790	0

Edges with identical switching-mask vectors enter every JC monomial together and are replaced by one effective product parameter. Permuting the two inheritance-choice bits and applying JC character automorphisms are exact open-cube symmetries.

The invariant deck consists of exact integer polynomials in the 15 nontrivial four-leaf JC orbit coordinates. The six core templates and their leaf-relabelling orbits distinguish equal-dimensional closures; a seventh multihomogeneous polynomial refines the directed signature universe. Each pullback is evaluated as a sparse polynomial over $\mathbb{Z}$. A zero bit means the pullback is the zero polynomial, not that it vanished at sampled points.

The containment orientation is as follows. If a source model were contained in a target model, then every polynomial vanishing on the target would vanish on the source. For unequal signatures we often obtain the stronger stochastic certificate used here: an invariant vanishes identically on the source and has one strict sign on the complete open target cube. That proves the two open images disjoint, and therefore excludes either orientation of open containment. For equal-dimensional irreducible closures, any invariant distinguishing the closures in either orientation is already sufficient to exclude containment.

7.2 Theta and cycle atlases

The canonical bounded counts are summarized in [table 2](#). These are counts of canonical signature records, not raw network presentations.

For theta-to-theta comparisons, equal signatures correspond exactly to the same canonical topology modulo T . At five outgoing ports, all 12,720 intersecting weak-target presentations are themselves strong and graphically replay to the source modulo T . At six outgoing ports the analogous complete replay checks 43,920 presentations. Every strict pair has an exact target pullback whose irreducible factors are one-signed on the open cube; tensor-product Bernstein coefficients provide an independent positivity certificate.

For cycles, the two-outgoing triangle is a single T class. Strong classes at three and four outgoing ports have 9 and 48 distinct signatures, respectively. Their weak atlases at sizes three through six have 12, 63, 390, 2,790 signatures. Equal signatures retain the selected path-sink child and replay to isomorphism or T ; there is no strict cycle-to-cycle pair. Strong theta sources and weak cycle targets have empty signature intersections at outgoing sizes three through six.

The only remaining direction is a cycle source against a weak theta target that mimics the cycle after some reticulation information is marginalized. Most such targets are separated after adding at most two rigid-support labels. Exactly 192 labelled core-3 presentations require three labels and hence seven outgoing ports.

Table 3: The eight seven-port role orbits. Each has all 24 permutations of the four original outgoing labels.

orbit id	selected path-sink child?	ordinary counts on (UV, SU, SV, UX, VX)
629	no	$(0, 0, 1, 1, 2)$
644	no	$(0, 1, 1, 2, 0)$
649	no	$(0, 2, 0, 1, 1)$
650	no	$(0, 2, 0, 2, 0)$
685	no	$(2, 1, 0, 1, 0)$
700	yes	$(0, 0, 2, 0, 1)$
705	yes	$(0, 1, 0, 2, 0)$
706	yes	$(0, 1, 1, 0, 1)$

7.3 The seven-port core-3 universe

Use the core-3 segment order

$$(UV, SU, SV, UX, VX),$$

with source S , tree branch U , branch reticulation V , and path-sink reticulation X . Its minimum repairs are

$$\{UV, VX\}, \quad \{UX, VX\}.$$

The residual records have the eight role patterns in [table 3](#).

Proposition 7.1 (Exhaustive seven-port universe). *The 192 residual labelled presentations are exactly the eight free S_4 orbits in [table 3](#). Conservative completion with labels 5, 6, 7 gives 1,686 distinct canonical labelled mixed graphs, each satisfying the S_{TC} local criterion. No residual relation is omitted or counted twice.*

Proof. The primitive core has four distinguished directed roles and therefore trivial directed automorphism group. A role pattern is determined by whether the sink child is selected and by the weak composition of the four ordinary selected labels over the five directed segments, subject to the condition that a full strong completion exists and that fewer than three additions do not contain a rigid support. Exhaustive solution of these finite constraints gives precisely the eight rows in [table 3](#). Since the core automorphism group is trivial, no nonidentity permutation of the four original labels fixes a presentation; every orbit has size 24 and the eight orbits are disjoint.

For each representative, assign the three new labels to all sink/repair placements that complete one of the minimum repairs, insert their port vertices, and canonicalize the resulting mixed graph. The local criterion of [theorem 2.4](#) is checked on the full completed graph. The primary and adversarial enumerators, written independently, obtain the same 1,686 canonical codes and the same complete displayed-tree monomial schemas. The certificate includes the code and schema digest for every record. $\square$

7.4 Trinet inequalities on restored labels

For three selected boundary ports a, b, c , let

$$F_{abc} = r_{ab}r_{ac}r_{bc} - u_{abc}^2, \tag{15}$$

where r_{ab}, r_{ac}, r_{bc} are the pair Fourier coordinates and u_{abc} is the all-distinct zero-sum coordinate. Englander et al. proved that this polynomial vanishes on a three-leaf tree and is strictly positive on every strict level-1 or level-2 JC trinet in their open domain [7, Proposition 2.26 in the April 2025 version]. We use the equivalent boundary-tensor normalization.

Lemma 7.2 (Cycle star pattern). *Let s be the path-sink label of a cycle factor. Then*

$$F_{abc} = 0 \iff s \notin \{a, b, c\}.$$

If the sink is selected, the pullback has one of the forms

$$F_{abc} = \lambda(1 - \lambda)M(1 - y)^2, \quad (16)$$

$$F_{abc} = \lambda(1 - \lambda)M(1 - yz)^2, \quad (17)$$

where M is a positive monomial and $y, z \in (0, 1)$ are path products. Thus $F_{abc} > 0$ whenever the sink is present.

Proof. If the sink is absent from the restriction, both cycle routes induce the same three-leaf tree after degree-two suppression, so (15) is the tree equation. If the sink is present, expand the two displayed-tree terms. When the two ordinary selected ports lie on one side, common pendant factors yield (16); when they lie on opposite sides, the intervening path contributes the product yz and yields (17). All removed factors are strictly positive. $\square$

On the original four outgoing labels plus the incoming boundary, the F zero pattern separates orbits 629, 644, 650, 685, 700, 706 for every possible source-sink label. Orbits 649 and 705 have the same cycle-star pattern on these five boundaries; the restored support labels are essential.

Lemma 7.3 (Orbit 649 separator). *Every strong completion of orbit 649 has a newly selected target sink $e \in \{5, 6, 7\}$. On the full target parameterization,*

$$F_{12e} = M_A(1 - \lambda_V)(1 - \lambda_X)(1 - a)^2 \times [\lambda_V(1 - \lambda_X)b + \lambda_Xc] > 0, \quad (18)$$

where M_A, a, b, c are positive products of open edge multipliers. The competing cycle has sink label 4 and hence $F_{12e} = 0$.

Lemma 7.4 (Orbit 705 separator). *Every strong completion of orbit 705 places a new label $e \in \{5, 6, 7\}$ on segment $V \rightarrow X$. Its full target parameterization satisfies*

$$F_{12e} = M_B\lambda_V(1 - \lambda_V)(1 - a)^2 > 0, \quad (19)$$

where $M_B, a > 0$ and $a < 1$. The competing cycle again has $F_{12e} = 0$.

Proofs of theorems 7.3 and 7.4. For each canonical completed target, marginalize the complete seven-port displayed-tree parameterization to $(1, 2, e)$ and expand (15). Group edge parameters having identical descendant masks into path products. Sparse exact factorization yields (18) and (19). Every factor is strictly positive on Θ_0 : inheritance probabilities and complements lie in $(0, 1)$, path products lie in $(0, 1)$, and the bracket in (18) is a positive sum. The independent reviewer reconstructs and expands all 486 reduced witness tensors occurring in these two orbits. $\square$

Theorem 7.5 (Seven-port separation). *Every residual cycle-source/core-3-theta-target relation has disjoint complete open JC stochastic images. In particular, none gives a lower-dimensional open intersection, one-sided containment, or full-dimensional overlap.*

Proof. Six of the eight role orbits are separated by the original five-boundary F pattern. The remaining two are separated by [theorems 7.3](#) and [7.4](#). The S_4 action transports the selected triples and formulas to all 192 labelled records. Since one polynomial is identically zero on the cycle source and strictly positive at every open target parameter, the open images are disjoint. $\square$

7.5 Local dimensions

For the incoming-port parameterizations in the seven-port certificate, the root is outside the local factor and the three-edge neighborhoods of distinct reticulations are disjoint. If there are n observed boundary leaves and r reticulations, binary degree counting gives

$$E = 2n + 3r - 2. \quad (20)$$

There are $E + r$ raw JC edge/inheritance parameters. One root-split gauge and two gauges per reticulation are unobservable. Locally, if the incoming multipliers are a, b , the outgoing multiplier is c , and the inheritance probability is λ , then a nonzero descendant character sees only

$$\lambda ac, \quad (1 - \lambda)bc,$$

while the zero character combines the two inheritance weights to one. Therefore

$$\dim \mathcal{V} \leq E + r - (1 + 2r) = 2n + 2r - 3. \quad (21)$$

For seven outgoing ports plus one incoming boundary, $n = 8$. The cycle has $r = 1$ and the theta target $r = 2$, giving upper bounds 15 and 17. Exact modular Jacobian minors are nonzero for all seven cycle side-count profiles and all 65 core-3 profiles appearing among the completions, so the bounds are attained over $\mathbb{Q}$; see [section A](#).

Theorem 7.6 (Complete nonroot local classification). *Let B, B' be arbitrary finite port-labelled nonroot S_{TC} level-2 local factors. Then*

$$B \preceq_{JC} B'$$

if and only if B and B' are port-labelled isomorphic or related by ordinary triangle redirection T . Every surviving relation is a symmetric regular overlap.

Proof. Choose a rigid support in the source. By [theorem 6.7](#), the corresponding target restriction occurs in the complete strong/weak bounded atlases. Theta/theta, cycle/cycle, and theta-source/cycle-target relations are already classified through the support bounds. The only cross-generator direction requiring a larger completion is the 192-record core-3 universe, closed by [theorem 7.5](#). Hence every source support, one-port probe, and two-port probe is target-equivalent only by isomorphism or T . The ordered-word reconstruction of [theorem 6.6](#) lifts these consistent bounded identifications to the complete arbitrary subdivisions.

Conversely, a port-labelled isomorphism is trivial. The common regular germ for T and its contextual gluing are proved in [section 8.3](#); they give symmetric full-dimensional overlap. $\square$

8 Root reduction and local-to-global completeness

The local theorem of [theorem 7.6](#) uses a distinguished incoming boundary. We first show that no new ambiguity arises when the root lies inside a local factor. We then combine cut preservation and analytic peeling to prove the global classification.

8.1 Rerooting through an ordinary tree-child path

Lemma 8.1 (Ordinary path to a leaf). *Let N^+ be a rooted tree-child partner of a standard semi-directed topology. Starting at the root, one can choose a child repeatedly to obtain a root-to-leaf path whose nonterminal vertices are ordinary tree vertices.*

Proof. At every internal vertex tree-childness supplies a child that is a tree vertex or a leaf. Choose such a child. Acyclic finiteness prevents repetition and forces termination at a leaf. No reticulation occurs on the chosen path. $\square$

Theorem 8.2 (Root reduction). *Every root-local factor of a standard semi-directed S_{TC} topology can be rerooted through an ordinary path onto a pendant edge so that it becomes a nonroot factor with a distinguished incoming boundary. The standard semi-directed topology and the complete open JC model germ are unchanged.*

Proof. Choose a rooted tree-child partner and an ordinary root-to-leaf path from [theorem 8.1](#). Suppress the old root, reverse the ordinary edges along the path, and insert a new binary root on the terminal pendant edge. Each internal path vertex exchanges one incoming and one outgoing ordinary edge, so its bidegree remains $(1, 2)$. Reticulation arrowheads are unchanged. A directed cycle in the new rooting would combine a reversed path interval with an old directed path returning to a higher path vertex, contradicting acyclicity of the original rooting. Thus the new rooting is admissible. Since the semi-directed topology lies in S_{TC} , the new rooting is tree-child.

Standard reduction forgets every reversed ordinary direction and suppresses the new root, so it returns the same mixed graph. Under JC, uniform stationarity and reversibility make the distribution of every displayed tree independent of root location. If rerooting splits an effective ordinary multiplier $x \in (0, 1)$, take

$$y = \frac{1+x}{2}, \quad z = \frac{2x}{1+x}, \quad (22)$$

so $0 < y, z < 1$ and $yz = x$. Conversely, suppressing the inserted root multiplies its two arm multipliers.

There is one retained-arrowhead case. Suppose the old root was inserted on an edge entering a reticulation. In a switching that selects this parent, the two root-arm multipliers occur consecutively and hence only through their product. In a switching that selects the other parent, the remaining one-child root stem subtends all observed leaves; its descendant character sum is zero, so its Fourier exponent is zero. Suppression therefore again preserves every term of [\(1\)](#). The parameter transformations are analytic and remain in the open cube, giving equality of the local model germs. $\square$

The incoming-boundary label used in the finite atlas is not observed. The atlas contains every relative ordering of boundary labels, so choosing another pendant root site merely permutes complete joint-tensor coordinates.

8.2 Localization of one-sided containment

We now make the projection step precise. Let p lie in a source-relative open set witnessing $N \preceq_{\text{JC}} N'$. By [theorem 5.5](#), both networks have the same labelled bridge tree $\mathcal{T}$. Apply [theorem 4.4](#) to a regular source point in the containment set. Let

$$\mathcal{U} = \prod_{v \in V(\mathcal{T})} U_v \times \prod_{e \in E(\mathcal{T})} I_e$$

be a source product chart, and write Γ for its analytic contraction into observed space.

The target model has the same cut factorizations at every common distribution. Anchor extraction, which depends only on the observed tensor, maps the target set into the same product of projective tensor spaces and bridge coordinates. It is not necessary to choose a regular target preimage.

Proposition 8.3 (Directed product localization). *Suppose a source-relative open subset of $\Gamma(\mathcal{U})$ is contained in the target model. After shrinking $\mathcal{U}$, for every component v the projection U_v contains a nonempty source-relative open subset of the local source model that lies in the corresponding local target model. Thus each local pair satisfies one-sided containment in the source-to-target direction.*

Proof. Because Γ is an analytic diffeomorphism on the source gauge slice, pull the containment set back to a full-dimensional semialgebraic subset $S \subset \mathcal{U}$. By [theorem 9.1](#), S contains a nonempty open box after shrinking around a smooth point:

$$B = \prod_v B_v \times \prod_e J_e \subseteq S.$$

For every $b \in B$, apply the observed anchor-extraction map to the target representation of $\Gamma(b)$. Uniqueness of positive rank-one factorization up to the already fixed reciprocal gauges implies that the extracted local tensor at v is exactly the v -coordinate of b . Hence B_v lies in the target local tensor set. Since B_v is open relative to the local source germ, it witnesses local one-sided containment. This argument uses no target regularity and no global selection of target parameters. $\square$

Corollary 8.4 (No cross-blob compensation). *A coordinated change in several distinct blobs cannot create a one-sided global containment unless every corresponding local source model is one-sided contained in its local target model.*

This is the precise sense in which observational ambiguity factorizes over blobs. Reciprocal bridge gauges do not permit one local defect to be cancelled by another: they have been fixed by the intrinsic positive anchor chart.

8.3 Sufficiency of embedded triangle redirection

Lemma 8.5 (Three ordinary external arms). *Every triangle in a valid binary standard semi-directed S_{TC} topology has exactly one reticulation, and each of its three external arms is an ordinary undirected edge.*

Proof. A triangle with no reticulation cannot occur in a rooted acyclic binary network blob. Two reticulations on a triangle would force either a reticulation-to-reticulation edge or a tree vertex tailing two retained arrows, violating [theorem 2.4](#). Hence there is one reticulation. The other two triangle vertices each tail one retained edge into it. The local strong criterion forces their remaining

incident external edge to be undirected. At the reticulation, the two triangle edges are its retained incoming edges. If its external edge were also retained, that reticulation would be the tail of an edge entering another reticulation and would have no incident undirected edge, again contradicting [theorem 2.4](#). Its external edge is therefore ordinary and undirected in the standard reduction. $\square$

Cut the three external arms beyond the triangle and let Q denote the normalized JC three-boundary tensor. The three orientations of the triangle define three maps into the same four-dimensional boundary-tensor space. The exact local certificate gives a common strict tensor point

$$r_{12} = r_{13} = r_{23} = \frac{7}{64}, \quad u_{123} = \frac{35}{1024},$$

and a rank-four parameter minor

$$-\frac{343}{16,777,216} \neq 0$$

for each orientation. Thus the three local images contain one common relative open tensor neighborhood $U \subset Q$.

Proposition 8.6 (Contextual T gluing). *Let two valid standard semi-directed networks differ only by an ordinary triangle redirection, with an otherwise identical context. Their JC model images share a full-dimensional regular germ, even if the context reconnects two of the triangle arms within the same level-2 blob.*

Proof. By [theorem 8.5](#), all three external arms are always-present ordinary JC edges. Cutting them factors either network parameterization as

$$\Theta_{\Delta} \times \Theta_{\text{ctx}} \longrightarrow Q \times \mathcal{C} \xrightarrow{H} \mathbb{R}^{4^{|X|}},$$

where the second map is one common analytic tensor contraction. Uniform stationarity changes between a normalized joint boundary tensor and a conditional port tensor only by orientation-independent powers of $1/4$, absorbed into H . The local triangle maps are submersions onto the same open set $U \subset Q$.

Choose a regular point of the generic contraction H over $U \times \mathcal{C}$. A maximal minor of DH cannot vanish on all of the real open set $U \times \mathcal{C}$ unless it is the zero polynomial; choose a point where it is nonzero. Composing with either local rank-four triangle map gives the same maximal rank, because both composites factor through $Q \times \mathcal{C}$ and no composite can exceed the rank of H . The inverse-function/constant-rank theorem gives a common full-dimensional regular image germ. No separation of the context into three independent pendant components is required. $\square$

8.4 Proof of the global theorem

Proof of theorem 3.2. Assume $N \preceq_{\text{JC}} N'$. By [theorem 5.5](#), their labelled reduced bridge trees agree. Extract projective local tensors and choose the analytic peeling chart of [theorem 4.4](#). Proposition 8.3 produces one-sided containment for each corresponding local factor. Apply root reduction if the global root is represented in a local factor, and then apply [theorem 7.6](#). Every local pair is port-labelled isomorphic or T-related.

Conversely, assume the bridge trees agree and every local pair has one of these two relations. Isomorphic factors are identified directly. For each redirected triangle, [theorem 8.6](#) gives a common

local regular germ in the complete surrounding context. Choose compatible positive anchor gauges and glue recursively along the bridge tree using (5)–(6). The source and target have the same total local-plus-bridge dimension, and the block-triangular peeling Jacobian is of full rank. Hence the global models share a symmetric full-dimensional regular germ. This proves both directions and the final assertion. $\square$

9 Genericity and reconstruction

The classification theorem is formulated through full-dimensional semialgebraic containment. We now connect this relation to the usual algebraic meaning of generic identifiability and give a constructive canonical reconstruction procedure.

9.1 Full dimension produces a relative open source set

We use standard facts from real algebraic geometry; see [2, 3]. All model images and intersections below are semialgebraic by Tarski–Seidenberg.

Lemma 9.1 (Semialgebraic interior). *Let M be a smooth d -dimensional semialgebraic manifold and let $S \subseteq M$ be semialgebraic. If $\dim_{\mathbb{R}} S = d$, then S contains a nonempty relatively open subset of M . Moreover, this subset may be chosen in the regular locus of any polynomial parameterization whose image contains a dense open subset of M .*

Proof. Take a finite semialgebraic Whitney stratification of S compatible with the singular/critical loci of the parameterization. Some stratum has dimension d . A d -dimensional embedded submanifold of the d -manifold M is relatively open near each of its points. The critical image has dimension less than d at a generically full-rank parameterization, so the top stratum may be chosen outside it. $\square$

Thus the definition of $N \preceq_{\text{JC}} N'$ is equivalent to saying that the intersection has the full real dimension of the source image. It is stronger than a single common point and weaker than equality of complete stochastic images.

9.2 Proper algebraic exceptional sets

Fix a topology N on n leaves. Let $\mathcal{N}_n$ be the finite set of valid labelled topologies in the theorem’s class. Finiteness follows from the tree-child reticulation bound. In a rooted binary network with r reticulations and n leaves, the number of tree vertices is $n + r - 2$ and the total number of vertices is $2n + 2r - 1$. Vertex-disjoint tree-child paths from the root and reticulations to leaves give $r \leq n - 1$, and hence at most $4n - 3$ vertices. Only finitely many labelled mixed graphs occur.

For every $N' \in \mathcal{N}_n$ not T-equivalent to N , define

$$E_{N,N'} = \overline{\mathcal{M}_N^{\text{JC}} \cap \mathcal{M}_{N'}^{\text{JC}}}^Z \subseteq \mathcal{V}_N^{\text{JC}}.$$

Proposition 9.2 (Properness of non-T intersections). *For every non-T topology N' , $E_{N,N'}$ is a proper algebraic subset of $\mathcal{V}_N^{\text{JC}}$.*

Proof. Suppose $E_{N,N'} = \mathcal{V}_N^{\text{JC}}$. The real semialgebraic intersection then has real dimension $\dim_{\mathbb{C}} \mathcal{V}_N^{\text{JC}}$: real and complex dimensions agree with the common maximal real/complex Jacobian rank, and complexification preserves the Krull dimension of the real Zariski closure. By [theorem 9.1](#), the

intersection contains a source-relative open regular subset, so $N \preceq_{\text{JC}} N'$. Theorem 3.2 then forces N' to be T-equivalent to N , a contradiction. $\square$

The finite union

$$E_N = \bigcup_{N' \not\sim_{\text{T}} N} E_{N,N'}$$

is therefore proper. Choose a polynomial f on observed space that vanishes on E_N but not identically on $\mathcal{V}_N^{\text{JC}}$. Dominance of the parameterization onto its closure makes $f \circ \phi_N$ a nonzero polynomial on parameter space. Its zero set is a proper algebraic subset. This proves theorem 3.3 in the stated parameter-space sense.

The exceptional locus also includes the finitely many rank/minor degeneracies needed by the constructive reconstruction. Adding these proper algebraic subsets does not change properness.

9.3 Canonical reconstruction procedure

We describe an exact algorithm for a generic distribution p known to arise from the class. It is structural rather than a search over all rooted DAGs.

- Step 1. Fourier transform.** Apply the invertible G^X Fourier transform to the pattern probabilities and retain the zero-sum coordinates.
- Step 2. Recover bridge splits.** For every nontrivial leaf split $A \mid B$, compute the rank of $\text{Flat}_{A|B}(p)$. By theorem 5.1, rank at most four is equivalent to a bridge split. Assemble the compatible split system into the labelled homeomorphism-reduced bridge tree.
- Step 3. Peel local tensors.** At each bridge and character sum, factor the positive rank-one Fourier block by the anchor formulas of theorem 4.3. Recursively peel the bridge tree, choosing the prescribed positive gauge slice. This yields one projective boundary tensor for every local factor and one bridge multiplier per edge.
- Step 4. Identify the local core.** Evaluate the exact bounded invariant deck on all restrictions through seven outgoing ports. Cycle/theta and strong/weak target signatures, together with the F_{abc} signs of theorem 7.5, identify the primitive core and a rigid support modulo T.
- Step 5. Recover segment words.** For each additional descendant block, use support-plus-one restrictions to locate its core segment. For each pair on one segment, use support-plus-two restrictions to determine their order. Sort by these comparisons to obtain the complete labelled word on each directed segment.
- Step 6. Canonicalize.** Form every structurally valid ordinary triangle redirection, retain only globally rootable S_{TC} mixed graphs, and compute their labelled canonical encodings. Return the lexicographically least encoding as the canonical representative modulo T. If requested, also return the complete structural T-equivalence class. This structural list is not asserted to equal the set of topologies whose stochastic images contain the particular input distribution p ; determining that pointwise list requires a separate semialgebraic membership test for each redirected endpoint.

Theorem 9.3 (Correctness of reconstruction). *Outside a proper algebraic exceptional subset of parameter space, the procedure terminates and returns the unique canonical semi-directed topology modulo labelled isomorphism and ordinary triangle redirection. Optionally, it returns the complete structural T-equivalence class of that canonical topology.*

Proof. There are finitely many splits and bounded restrictions, so the procedure terminates. Steps 2 and 3 are correct by [theorems 4.4 and 5.1](#). The complete local atlas and bounded-support theorem prove Steps 4 and 5 and identify one structural topology class modulo T . Step 6 performs a finite graph-theoretic normalization inside that class. [Theorem 3.2](#) excludes every non- T topology from full-dimensional generic compatibility, which proves uniqueness of the canonical quotient representative. [Proposition 8.6](#) shows only that valid redirected endpoints share a full-dimensional regular germ; it does not imply that the particular input point p belongs to every redirected model. The exceptional set is the finite union of the proper rank, invariant, and non- T intersection loci described above. $\square$

Remark 9.4 (Structural versus pointwise triangle compatibility). *The reconstruction theorem identifies the structural quotient by T . It does not solve the distinct pointwise problem of deciding, for a fixed exact distribution p , which orientations in that structural class actually contain p in their open stochastic images. The three triangle models can have both full-dimensional intersections and full-dimensional differences, so that pointwise question requires additional semialgebraic membership tests.*

We do not claim a polynomial-time bit-complexity bound. Direct split testing is exponential in n , and exact algebraic-number arithmetic may dominate practical cost. The result is nevertheless constructive and avoids enumerating all rooted DAGs on up to $4n - 3$ vertices.

10 Sharpness: a non-triangle ambiguity in $W_{\mathsf{TC}} \setminus S_{\mathsf{TC}}$

We now prove [theorem 3.5](#) without appealing to any result outside this paper and its exact supplement. The construction is a pair of four-leaf level-2 theta networks whose pendant placements differ. Their standard semi-directed topologies are weakly tree-child but not strongly tree-child, and their JC images meet along a regular eight-dimensional open set.

10.1 The four-leaf pair

Let the internal vertices be a root ρ , tree vertices A, B, D, E , and reticulations C, F . Both rooted networks have internal arcs

$$\rho \rightarrow A, \quad \rho \rightarrow C, \quad A \rightarrow B, \quad B \rightarrow C, \quad C \rightarrow D, \quad D \rightarrow E, \quad A \rightarrow F, \quad E \rightarrow F. \quad (23)$$

Their pendant arcs are

$$N : \quad B \rightarrow 1, \quad D \rightarrow 2, \quad F \rightarrow 3, \quad E \rightarrow 4, \quad (24)$$

$$N' : \quad E \rightarrow 1, \quad D \rightarrow 2, \quad F \rightarrow 3, \quad B \rightarrow 4. \quad (25)$$

After suppressing the root artifact, the nontrivial blob is a theta graph with three A – C paths

$$A - C, \quad A - B - C, \quad A - F - E - D - C.$$

Its three simple cycles have lengths 3, 5, 6, so both networks are binary and level 2, and each blob contains exactly one triangle. The two standard reductions are shown in [figure 8](#).

Proposition 10.1 (Tree-child class and topological separation). *The supplied rooted representatives of N and N' lie in R_{TC} . Each standard semi-directed topology has five admissible rootings, exactly two of which are tree-child. Hence*

$$N, N' \in W_{TC} \setminus S_{TC}.$$

The two standard semi-directed topologies are nonisomorphic and are not related by ordinary triangle redirection.

Proof. In the rooted networks (23)–(25), every internal vertex has a tree-or-leaf child and neither reticulation has a reticulation child. Thus both chosen rootings are tree-child.

For the standard reductions, retain the arrowheads entering C and F , suppress ρ , and enumerate each edge on which a root can be inserted without changing the retained arrowheads. Propagating the forced ordinary orientations from the proposed root gives exactly five admissible rootings. The root sites $A - C$ and $A - F$ give tree-child rootings; the remaining three give a tail of a retained arrow two reticulation children. The same count and the same two root sites occur for both pendant placements. This proves membership in $W_{TC} \setminus S_{TC}$. The enumeration is reproduced from the arc lists by the independent checker in the supplement.

The unique triangle is $A - B - C$. In N , leaf 1 is adjacent to B , a triangle vertex; in N' , leaf 1 is adjacent to E , which is not on a triangle. This labelled graph property is preserved by isomorphism and by a redirection confined to the unique triangle. Therefore the two topologies are neither isomorphic nor T-equivalent. $\square$

10.2 Exact common JC point

Write every edge multiplier as $x_e \in (0, 1)$ and let $\lambda_C, \lambda_F \in (0, 1)$ be the two inheritance probabilities. It is convenient to suppress the root and use the effective product on the root-split $A - C$ edge. A rooted realization is recovered by the strict factorization

$$x_{\rho A} = \frac{2}{3}, \quad x_{\rho C} = \frac{3}{4}, \quad x_{\rho A} x_{\rho C} = \frac{1}{2}. \quad (26)$$

For N , take

$$\begin{aligned} x_{D2} &= \frac{1}{2}, & x_{DE} &= \frac{2}{5}, & x_{E4} &= \frac{3}{8}, & x_{EF} &= \frac{1}{3}, & x_{F3} &= \frac{1}{2}, \\ x_{CD} &= \frac{9}{20}, & x_{AB} &= \frac{3}{5}, & x_{B1} &= \frac{1}{5}, \\ x_{AC} &= x_{AF} = x_{BC} = \frac{1}{2}, & \lambda_C &= \lambda_F = \frac{1}{2}. \end{aligned} \quad (27)$$

Let β be the smaller real root of

$$43337075\beta^2 - 36083110\beta + 7336259 = 0, \quad \frac{441}{1250} < \beta < \frac{3529}{10000}. \quad (28)$$

For N' , take

$$\begin{aligned}
x_{D2} &= \frac{1}{2}, & x_{DE} &= \frac{171}{775}, & x_{AF} &= \frac{10339}{53010\beta}, \\
x_{AB} &= \frac{24835\beta}{20678 - 24835\beta}, & x_{F3} &= \frac{1767}{4832}, & x_{CD} &= \frac{9934}{12215}, \\
x_{E1} &= \frac{31}{190}, & x_{B4} &= \frac{3}{20\beta}, \\
x_{AC} &= x_{EF} = x_{BC} = \frac{1}{2}, & \lambda_C &= \lambda_F = \frac{1}{2}.
\end{aligned} \tag{29}$$

The isolating interval in (28) proves directly that every quantity in (29) lies strictly between zero and one.

The four-leaf JC orbit space has fourteen nonconstant coordinates. In the ordering

$$(A, B, C, D, E, F, G, H, J, K, L, M, N, O)$$

used in section B, the common point is

$$(A, \dots, H) = \left(\frac{277}{8000}, \frac{3}{40}, \frac{67}{2400}, \frac{127}{16000}, \frac{27}{5000}, \frac{81}{5000}, \frac{153}{160000}, \frac{9}{500} \right), \tag{30}$$

$$(J, \dots, O) = \left(\frac{27}{16000}, \frac{261}{320000}, \frac{27}{10000}, \frac{27}{20000}, \frac{153}{320000}, \frac{183}{80000} \right). \tag{31}$$

Proposition 10.2 (Exact Fourier equality). *Substitution of (27) and (29) into the displayed-tree formula (1) gives identical values for all 256 four-leaf Fourier coordinates, and hence for all 256 pattern probabilities.*

Proof. Coordinates with nonzero total character sum vanish on both sides. The remaining coordinates reduce, under JC character symmetry, to the fourteen entries in (30)–(31). For the source, direct rational substitution gives those values. For the target, clear the denominators and reduce every numerator difference modulo the quadratic polynomial in (28); each remainder is zero. The complete list of orbit representatives and the exact sparse reductions appear in section B, and the independent verifier evaluates the unreduced 256-coordinate tensors directly. The inverse Fourier transform is linear and invertible, so the pattern distributions agree as well. $\square$

10.3 A common regular eight-dimensional germ

The equality at one point is not enough for the sharpness theorem. We now show that the intersection contains a full-dimensional regular neighborhood.

Both parameterizations satisfy the six equations

$$J - K - M + N = 0, \tag{32}$$

$$J - AH - BF + CE = 0, \tag{33}$$

$$GL - EN = 0, \tag{34}$$

$$L^2 - BEH = 0, \tag{35}$$

$$BM - DL - B^2F + BCE = 0, \tag{36}$$

$$BEO - BGH - CEL + DEH = 0. \tag{37}$$

On the physical locus $BE > 0$, five of these reconstruct J, K, M, N, O from $(A, B, C, D, E, F, G, H, L)$,

while (35) gives

$$L = +\sqrt{BEH}.$$

Thus the physical main locus is a smooth sheet of the irreducible variety with localized coordinate ring

$$\mathbb{C}[A, B, C, D, E, F, G, H, L, (BE)^{-1}]/(L^2 - BEH), \quad (38)$$

which has dimension eight. Smoothness on the open locus follows from $\partial(L^2 - BEH)/\partial H = -BE \neq 0$.

For the source, fix

$$x_{AC} = x_{AF} = x_{BC} = \lambda_C = \lambda_F = \frac{1}{2}$$

and use the eight variables

$$(P, s, Q, t, R, u, v, S) = (x_{D2}, x_{DE}, x_{E4}, x_{EF}, x_{F3}, x_{CD}, x_{AB}, x_{B1}).$$

The Jacobian determinant of the map to $(A, \dots, H)$ is

$$-\frac{P^3 s^3 Q^4 t R^4 u^3 v S^3 (s-1)^2 (v-1)(v+1)^2}{16384}, \quad (39)$$

which is nonzero at (27). For the target, fix

$$x_{AC} = x_{EF} = x_{BC} = \lambda_C = \lambda_F = \frac{1}{2}$$

and use

$$(P', x, y, z, R', w, S', Q') = (x_{D2}, x_{DE}, x_{AF}, x_{AB}, x_{F3}, x_{CD}, x_{E1}, x_{B4}).$$

The corresponding determinant is

$$-\frac{(P')^3 x^2 y^2 z (R')^4 w^4 (S')^3 (Q')^4 (x-1)^2 (z-1)(z+1)^3}{32768}, \quad (40)$$

which is nonzero throughout the open target point (29).

Corollary 10.3 (Four-leaf regular overlap). *The two four-leaf JC images share a relatively open regular eight-dimensional subset. In particular, $N \bowtie_{\text{JC}} N'$.*

Proof. The six identities place both images in the same irreducible eight-dimensional locus (38). Equations (39) and (40) show that each image is locally an open eight-dimensional submanifold of the positive smooth sheet at their common point. Their two image neighborhoods therefore have a nonempty relative intersection in that sheet. $\square$

10.4 The extra parameter defect

The eight-dimensional result has one more generic fiber dimension than the standard local edge gauges alone predict. This point matters for the all- n dimension formula.

A rooted binary network with n leaves and r reticulations has

$$E = 2n + 3r - 2 \quad (41)$$

network edges; see section A. For $r = 2$, the raw JC parameter count is $E + r = 2n + 6$. The ordinary

root-split gauge and two local gauges at each reticulation account for five generic fiber dimensions, suggesting the upper value $2n + 1$. For the Theta topology there is one further generic dependence. At $n = 4$, the exact implicit description (38) and the nonzero determinants above prove dimension eight rather than nine. Equivalently, the generic fiber has dimension six rather than five. This extra one-dimensional defect persists under the leaf substitutions below because each substitution contributes two new identifiable arm multipliers and does not interact with the original fiber.

10.5 Leaf substitution and the all- n family

Let P be the Fourier tensor of either member of the four-leaf pair, written with one selected leaf port a . Replace a by a cherry with new leaves 1, 2 and edge multipliers $u, v \in (0, 1)$. If g_X denotes the characters on all other leaves, then

$$\tilde{P}(g_X, g_1, g_2) = P(g_X, g_1 \oplus g_2) u^{\mathbf{1}_{[g_1 \neq 0]}} v^{\mathbf{1}_{[g_2 \neq 0]}}. \quad (42)$$

For any nonzero $h \in G$, positivity gives

$$uv = \tilde{P}(0, h, h), \quad (43)$$

$$\frac{u}{v} = \frac{\tilde{P}(g_X, h, 0)}{\tilde{P}(g_X, 0, h)} \quad \text{whenever } \bigoplus g_X = h. \quad (44)$$

Hence u and v are recovered uniquely by the positive square root, after which P is recovered from (42). Leaf substitution is therefore a positive analytic embedding with a positive analytic inverse onto its image. Replacing corresponding leaves of N and N' by the same rooted binary tree preserves the common tensor equality and adds exactly two model dimensions for every new leaf.

The standard semi-directed graphs remain nonisomorphic and non-T-equivalent because restricting to one chosen descendant leaf in every substituted component recovers the original labelled pair. A tree-child rooting extends through every substituted rooted tree, while each original non-tree-child admissible rooting remains non-tree-child after substitution. Thus membership in $W_{\text{TC}} \setminus S_{\text{TC}}$ is preserved.

Starting from the eight-dimensional four-leaf pair and applying $n - 4$ cherry substitutions yields

$$\dim \mathcal{M}_{N_n}^{\text{JC}} = \dim \mathcal{M}_{N'_n}^{\text{JC}} = 8 + 2(n - 4) = 2n.$$

The analytic inverse shows that the common open germ also gains two dimensions at every step. This proves [theorem 3.5](#) for every $n \geq 4$.

11 Computer-assisted proof and reproducibility

The proof combines conceptual arguments valid for arbitrary networks with exact finite calculations. This section describes where computation enters, why the finite universes are exhaustive, and how the publication archive checks the calculations independently. The integrity hashes distributed with the archive detect accidental file changes; they are not mathematical evidence. The mathematical evidence consists of regenerated graphs, exact polynomial identities, sign certificates, and nonzero Jacobian minors.

11.1 Division between written proof and finite verification

The following arguments are entirely symbolic and are proved in the body of the paper: Fourier bridge factorization and its reciprocal gauge group; analytic bridge peeling; the two-active-endpoint cut contradiction; reduction of level-2 blobs to a cycle or theta core; the automatic exclusion of multi-triangle weakly tree-child blobs; rigid-support and ordered-word reconstruction; root reduction; semialgebraic localization; and the local-to-global product theorem.

Finite computation is used for five tasks.

- (i) It independently enumerates every binary acyclic rooting of the sole two-triangle level-2 core, K_4 minus an edge, as an audit of [theorem 3.1](#).
- (ii) It enumerates all oriented cycle and theta cores satisfying the binary and S_{TC} conditions, and all bounded port placements required by the rigid-support theorem.
- (iii) It evaluates exact JC Fourier tensors and exact polynomial invariants on every canonical type, including every directed source–target comparison.
- (iv) It proves strict open-cube signs by sparse factorization or tensor-product Bernstein coefficients.
- (v) It certifies generic model dimensions by exact nonzero Jacobian minors, sometimes first located modulo a prime and then replayed exactly over $\mathbb{Q}$.

No floating-point fit is used to certify an overlap, containment, separation, sign, dimension, or graph count.

11.2 Primitive graph generation

The regenerative source begins with the definitions of the two branch vertices, source event, and reticulation events on three theta paths. It enumerates all placements and path orientations, imposes the required bidegrees, acyclicity, reachability, and the local S_{TC} criterion, and then quotients by branch reversal and path permutation. This gives four oriented theta cores. The cycle core is generated separately. A separate primitive program starts from the five edges of K_4 minus an edge, ranges over every reticulation pair and every internal or external binary root site, and rejects all 25 valid acyclic presentations as non-tree-child; an independently written C++ program repeats that census without importing the atlas code. Ordinary port vertices are inserted as ordered words on directed core segments. Minimum tree-child repairs are generated as minimal hitting sets of the local child obligations, rather than read from a frozen list.

For each bounded support size, canonical mixed-graph codes are computed by colour refinement followed, when necessary, by exact individualization. Taxon labels and the incoming-port role are included in the colours. Records are inserted into a dictionary keyed by the complete canonical code, which proves duplicate removal by construction. A second implementation uses an independent recursive canonicalization and compares the complete set of codes and orbit sizes.

The resulting structural counts are listed in [table 4](#). These are outputs of the primitive generators, not assumptions supplied to them.

The 192 residual records are regenerated as eight role patterns with a free S_4 action. For each record, all legal assignments of the three restored support labels are generated, and only standard S_{TC} completions are retained. Canonicalization produces the 1,686 completed mixed graphs used in [theorem 7.5](#). The independent reviewer starts again from the primitive core and verifies the orbit decomposition, class membership, canonical codes, and the complete displayed-tree monomial schemas.

Table 4: Finite local universes regenerated in the publication archive.

Universe	Sources	Targets or signatures	Strict directed cases
Two-triangle (1, 2, 2) theta	25 rooted presentations	7 symmetry orbits	0 tree-child survivors
Theta, five outgoing ports	8,520	16,590	18,480
Theta, six outgoing ports	10,980	218,925	21,960
Strong cycle, three outgoing ports	9	12 weak signatures	none survive
Strong cycle, four outgoing ports	48	63 weak signatures	no non-T survivor
Weak cycle, five outgoing ports	—	390	no theta-source intersection
Weak cycle, six outgoing ports	—	2,790	no theta-source intersection
Residual core-3 completion	192 role records	1,686 completed graphs	all 192 separated
Cut endpoints	177 types	453 one-active types	421 strict minors

11.3 Exact Fourier and invariant arithmetic

For every rooted realization and every reticulation parent choice, the software deletes the unselected incoming arcs and computes the descendant taxon set of each selected edge. Formula (1) is then evaluated as a sparse polynomial over $\mathbb{Z}$ or $\mathbb{Q}$. Marginal tensors are obtained by setting omitted leaf characters to zero, not by numerical summation.

On a four-port restriction, an edge is represented by the tuple of its descendant masks over all parent choices. Edges with identical tuples enter every JC monomial together and are replaced by the product of their multipliers. This produces an exact reduced tensor type. The finite atlas evaluates a symmetry-complete deck of integer invariants on each type. The source/target orientation is recorded explicitly. For a directed containment $\mathcal{M}_{\text{source}} \subseteq \mathcal{M}_{\text{target}}$, an invariant vanishing on the source and strictly nonzero on the target proves that the open images are disjoint; equal-dimensional closure separation also excludes containment by irreducibility.

The final seven-port calculation does not rely on a zero-signature heuristic. It expands the complete marginal of each restored label and verifies the factorizations (18) and (19) directly from the seven-port displayed-tree formulas.

11.4 Strict signs

A factor such as x , $1 - x$, $1 - xy$, or a positive convex combination has an immediate strict sign on $(0, 1)^d$. More complicated multivariate factors are converted exactly to the tensor-product Bernstein basis on $[0, 1]^d$. If all coefficients have one sign and at least one is nonzero, the polynomial has that strict sign on the open cube after any explicitly removed positive factors are restored. Coefficients are stored as arbitrary-precision integers or rational numbers.

The pointwise cut calculation regenerates 177 endpoint types and 453 one-active crossing types. An independent reviewer recomputes 547 sparse Bernstein expansions without importing the primary sign routine. It verifies all 151/26 endpoint dichotomy cases, all 421 strict wrong-split minors, and all 32 rank-one true-bridge cases. The two-active contradiction is then checked as a formal identity in the nine variables of theorem 5.4.

11.5 Exact dimensions and modular discovery

Jacobian rank is a generic algebraic property. The archive constructs the symbolic Jacobian of a chosen output coordinate block. Some useful minors were located by evaluation modulo the prime 1,000,003. A determinant that is nonzero modulo a prime is the reduction of a nonzero integer polynomial, and therefore proves a nonzero minor over $\mathbb{Q}$. The publication certificate stores the selected rows, columns, and modular value, and the independent checker re-evaluates each determinant from the primitive parameterization. The structural gauge upper bounds in [section A](#) then prove equality with the claimed model dimensions.

For the Theta sharpness pair, the stronger rational factorizations (39) and (40) are checked symbolically. The target arithmetic is carried out in the exact quadratic field defined by (28).

11.6 Independent replay and release commands

The archive contains separate primary and adversarial implementations for the automatic triangle bound, the seven-port closure, root reduction, pointwise cut theorem, final synthesis, and publication preflight. The review implementations do not import the primary graph generators or sign routines. They reconstruct the relevant objects from primitive encodings and compare complete canonical-code or polynomial sets, not only totals.

From the archive root, the short verification is

```
python3 verify_release.py
```

and the complete adversarial replay is

```
python3 verify_release.py --full-adversarial
```

The publication wrapper additionally regenerates the bounded graph universes, builds the manuscript, runs bibliography and PDF preflight checks, and verifies the theorem-to-certificate crosswalk. Pinned environment files and clean-shell commands are included for both Conda and standard virtual environments. Measured runtime and peak memory for the distributed environment are reported in `RUNTIME.md`.

11.7 Integrity and archival practice

The archive includes a SHA-256 manifest covering source, certificates, figures, manuscript files, and verification transcripts. The verifier checks that no required path is missing or duplicated and that generated certificates match the distributed versions. Hashes serve only as integrity controls. A mathematical claim is accepted by the release wrapper only after the corresponding generator, exact identity or sign check, and independent review assertion all pass. The detailed correspondence is given in [section C](#).

12 Discussion, limitations, and declarations

12.1 What exact JC data recover

The classification gives a concrete answer to the biological identifiability question within the stated class. An exact generic site-pattern distribution determines the homeomorphism-reduced arrangement of reticulated components, whether each local component is a cycle or a theta blob, the

placement of every labelled descendant block, and its position in the ordered port words. Ordinary triangle redirection is the only topology change that can remain observationally compatible on a full-dimensional model region. Thus the presence and location of a close reticulation can be identifiable even when its direction inside a three-cycle is not generically determined modulo that move.

This conclusion is stronger than algebraic distinguishability alone. The theorem classifies one-sided full-dimensional stochastic containment on the open biological parameter domain. It therefore excludes both equal-dimensional non- $\mathbb{T}$ overlap and a lower-dimensional topology whose generic distributions could all be explained by a larger competing topology. Conversely, the three ordinary triangle orientations are shown to share a regular open germ; equality of their complete stochastic images is neither asserted nor needed. In particular, the structural reconstruction returns a canonical topology modulo $\mathbb{T}$, not the input-specific list of every redirected orientation whose stochastic image contains the observed distribution.

The sharpness result explains why the strong convention is substantive. A topology in W_{TC} may admit a biologically appealing tree-child rooting while also admitting rootings that expose a vertex with two reticulation children. The Theta pair shows that this extra flexibility permits an exchange of labelled descendant components to become observationally invisible on a full-dimensional set. Requiring every admissible rooting to be tree-child removes precisely this mechanism in the theorem’s class.

12.2 Why no triangle-count hypothesis remains

The final theorem covers all binary standard semi-directed S_{TC} level-2 networks. No additional hypothesis on the number of triangles is hidden in the finite atlas. The reason is structural: a level-2 blob with two triangles must be the $(1, 2, 2)$ theta, or K_4 minus an edge, and [theorem 3.1](#) shows that this graph has no tree-child rooting. In fact, the exclusion already holds in the larger class W_{TC} . The four oriented theta cores and the cycle core therefore exhaust every local factor that can occur in the positive theorem.

The theorem is specific to the displayed-tree JC model with uniform root distribution, strictly positive finite branch lengths, and inheritance probabilities strictly between zero and one. Boundary parameters can collapse edges, remove reticulation choices, or create containments absent from the open model. Models with coalescent processes, site-rate variation, nonstationary substitution, or richer group-based edge parameters require separate analysis. Level greater than two also permits generator cores and triangle interactions not covered by the present structural lemma.

12.3 Statistical and computational limitations

Generic identifiability is an infinite-data property. It does not imply that all identifiable features can be estimated accurately from finite sequences, especially near the algebraic exceptional locus or near a triangle model intersection. The semialgebraic work on three-leaf triangles shows that substantial full-dimensional overlap may coexist with regions where the triangle direction is distinguishable. Developing statistically stable estimators for the recoverable canonical class is an important next step.

The reconstruction theorem is constructive but not offered as an efficient implementation. Its optional enumeration is the structural $\mathbb{T}$ -equivalence class; deciding which of those orientations contain one particular distribution is a separate semialgebraic membership problem. Testing all leaf splits directly is exponential, and exact algebraic arithmetic is expensive. The finite atlas suggests

practical invariant screens, but a statistically and computationally optimized inference method lies outside the present scope.

12.4 Open problems

Natural questions remaining within phylogenetic algebraic statistics include the following. Does an analogous canonical classification hold at level greater than two? Can the generic bridge and local-tensor reconstruction be made polynomial-time under an oracle model for exact probabilities? Which parts of the theorem survive under rate variation or nonstationary models? Finally, can the global bridge/local-tensor extraction be combined with the known semialgebraic triangle inequalities to determine, for one exact distribution, precisely which redirected orientations are pointwise compatible and to give a statistically useful confidence region for the canonical topology class?

Acknowledgements

The author thanks the authors of the cited phylogenetic-network identifiability papers for making their preprints and computational materials publicly available. The author also thanks the adversarial AI-assisted review processes and separately implemented automated checks that exposed and corrected several earlier incomplete arguments before this version was prepared. No institutional funding supported this work.

Use of artificial intelligence

OpenAI ChatGPT, including GPT-5.6 Pro, and Anthropic Claude were used during exploration and revision. These systems generated candidate arguments, symbolic-computation code, manuscript drafts, and adversarial review prompts. Distinct implementations were used to rederive the load-bearing finite calculations, and failed or incomplete intermediate claims were retained in the archived research logs. Artificial-intelligence systems are not authors. The human author personally reviewed the final mathematical statements, citations, manuscript, and release artifacts and accepts full responsibility for their accuracy. Prompts, logs, source code, exact certificates, and verification transcripts are archived with the reproducibility package.

Author contributions

Alec Kriebel conceived the project, selected and refined the research question, directed the computational and proof audits, checked the final arguments and citations, prepared the release, and accepts responsibility for the work in its entirety.

Competing interests

The author declares no competing interests.

Data and code availability

No empirical biological data were used. Complete LaTeX and TikZ source, primitive graph generators, exact Fourier and polynomial verifiers, finite certificates, independent replay programs, environment specifications, manifests, and successful clean verification transcripts accompany this manuscript. The versioned supplementary archive accompanying this submission is self-contained

and includes its own SHA-256 manifest and verification transcripts. A persistent repository identifier should be inserted in the journal metadata and this statement after the author deposits the final, checksum-matched archive. No persistent identifier is claimed in the present local release candidate.

A Dimension and gauge calculations

This appendix collects the dimension bounds used in the finite local atlases and explains how exact Jacobian minors certify equality.

A.1 Binary network counts

Let a rooted binary network have n leaves, r reticulations, and t tree vertices other than the root. Summing indegrees and outdegrees gives

$$E = t + 2r + n = 2 + 2t + r.$$

Hence

$$t = n + r - 2, \quad E = 2n + 3r - 2.$$

The raw JC parameter count is $E + r = 2n + 4r - 2$.

For a lifted incoming-port local factor, the root lies outside the blob and the two root arms enter only through one product, giving one root-split gauge. Suppose in addition that the three-edge neighborhoods of the reticulations are disjoint, as they are in the bounded cycle and theta parameterizations used here. At a reticulation with incoming multipliers a, b , outgoing multiplier c , and inheritance λ , the local contribution for a nonzero descendant character is determined by

$$\lambda ac, \quad (1 - \lambda)bc.$$

For zero descendant character the two parent choices combine to one. Thus the four raw local parameters contain two independent gauge directions. The resulting upper bound is

$$\dim \mathcal{V}_N^{\text{JC}} \leq E + r - (1 + 2r) = 2n + 2r - 3. \quad (45)$$

For the seven-outgoing comparison, the incoming port makes $n = 8$. The cycle has $r = 1$ and therefore dimension at most 15; the core-3 theta has $r = 2$ and dimension at most 17.

The disjoint-neighborhood hypothesis in this counting lemma is not asserted for every possible level-2 parameterization. When it does not apply, dimensions are computed from an explicit gauge chart or from the complete symbolic Jacobian. The global theorem itself needs only the local dimensions certified in the finite atlas.

A.2 Jacobian certification

Let $J(\theta)$ be an $m \times p$ Jacobian matrix with entries in $\mathbb{Z}[\theta]$ after clearing denominators. If a $d \times d$ minor has a nonzero value modulo a prime at some parameter assignment, its determinant is a nonzero polynomial over $\mathbb{Z}$ and hence over $\mathbb{Q}$. Therefore the generic rank is at least d . Combined with a structural upper bound such as (45), this proves the exact model dimension.

The seven-port certificate stores one selected minor for each of seven cycle side-count profiles and each of sixty-five core-3 theta count profiles represented among the 1,686 completions. The primary

and independent implementations reconstruct the parameterizations from the primitive graphs and re-evaluate all stored minors modulo 1,000,003. The resulting ranks are 15 and 17, respectively.

For the cut theorem, dimensions are not inferred from sampled rank. Exact sparse determinant factorizations or tensor-product Bernstein signs prove that the selected wrong-split minor is nonzero at every point of the open cube. For the Theta sharpness pair, equations (39) and (40) are exact rational-function factorizations and prove rank eight directly.

A.3 The Theta defect

For the four-leaf Theta pair, $n = 4$ and $r = 2$, so $E = 12$ and the raw parameter count is 14. The root-split plus two gauges per reticulation account for five dimensions of the generic fiber, leaving the naive upper count nine. The actual model has dimension eight. The six identities (32)–(37) identify its localized main locus with (38), while the nonzero rank-eight determinants prove that the image is dense there. Thus the generic fiber has dimension six: one more than the standard gauge count.

A cherry substitution adds two edge multipliers. Equations (43)–(44) recover both, so it adds exactly two image dimensions and no new fiber defect. Induction therefore gives dimension $8 + 2(n - 4) = 2n$ for the all- n sharpness family.

B Exact data for the Theta sharpness pair

This appendix records the exact coordinate convention and certificates used in [section 10](#). It is included so that the common point and regularity can be checked independently of the machine-readable files.

B.1 Fourier orbit coordinates

Identify the nonzero elements of $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ with 1, 2, 3. The fourteen nonconstant JC coordinate orbits are represented by

$$\begin{aligned} A &= r_{34}, \quad B = r_{24}, \quad C = r_{23}, \quad D = t_{234}, \quad E = r_{14}, \quad F = r_{13}, \quad G = t_{134}, \quad H = r_{12}, \\ J &= u_{1111}, \quad K = u_{1122}, \quad L = t_{124}, \quad M = u_{1212}, \quad N = u_{1221}, \quad O = t_{123}. \end{aligned}$$

Here r_{ij} denotes a two-leaf nonzero-character coordinate, t_{ijk} a three-leaf zero-sum coordinate, and u a four-leaf zero-sum coordinate in the indicated character orbit. Character relabelling supplies all remaining zero-sum coordinates.

At the common point, the values are those in (30)–(31). Direct substitution verifies (32)–(37). On $BE \neq 0$ they reconstruct

$$\begin{aligned} J &= AH + BF - CE, \\ N &= GL/E, \\ M &= (DL + B^2F - BCE)/B, \\ K &= J + N - M, \\ O &= (BGH + CEL - DEH)/(BE), \end{aligned}$$

with $L^2 = BEH$. This proves the coordinate-ring description (38).

B.2 Exact number field and positivity

The target field is $\mathbb{Q}(\beta)$, where β satisfies (28). Evaluating the polynomial at the two rational endpoints gives opposite signs, and its discriminant is positive, so the interval isolates the smaller real root. Rational interval arithmetic applied to each expression in (29) gives

$$0 < x_e < 1, \quad 0 < \lambda_C, \lambda_F < 1.$$

The smallest margins are retained in the exact certificate; no floating-point rounding is used.

For each zero-sum assignment $g \in G^4$, the source value is rational. The target value is represented as

$$\frac{a_g + b_g \beta}{d_g}, \quad a_g, b_g, d_g \in \mathbb{Z},$$

after reduction modulo (28). The verifier checks equality with the source by confirming $b_g = 0$ and a_g/d_g equal to the source value for all 64 zero-sum assignments, and then repeats the calculation on all 256 assignments, including the forced zeros.

B.3 Jacobian blocks

The source output block is (A, B, C, D, E, F, G, H) and the source input block is

$$(x_{D2}, x_{DE}, x_{E4}, x_{EF}, x_{F3}, x_{CD}, x_{AB}, x_{B1}).$$

After fixing the five parameters stated before (39), its determinant factors exactly as (39). The target uses the same output block and the input block

$$(x_{D2}, x_{DE}, x_{AF}, x_{AB}, x_{F3}, x_{CD}, x_{E1}, x_{B4}),$$

with determinant (40). Every factor is nonzero at the certified point. These are ordinary derivatives of the full displayed-tree Fourier parameterizations; they are not determinants of an implicit reconstruction map.

B.4 Admissible rootings

The standard mixed graph has retained arrowheads on $A \rightarrow C$, $B \rightarrow C$, $A \rightarrow F$, and $E \rightarrow F$. The independent rooting checker inserts a root on every compatible edge, orients all undirected edges away from it subject to the retained arrowheads, rejects cycles and incorrect bidegrees, and reduces the result back to the original mixed graph. Five rootings survive. Root insertions on $A - C$ and $A - F$ are tree-child; the other three are not. The result is identical for the two pendant placements because rooting depends only on the internal mixed graph.

C Finite-atlas generation and theorem-to-certificate crosswalk

The publication archive is organized so that every finite claim in the paper has a primitive generator, a mathematical certificate, and, where load-bearing, an independent replay. The principal correspondence is listed below. Paths are relative to the archive root; the machine-readable version is `THEOREM_CERTIFICATE_CROSSWALK.md`.

Automatic triangle bound. Primary: `publication/src/verify_multitriangle_exclusion.py`. Independent review: `publication/review/review_multitriangle_exclusion.cpp`. Output: `publication/certificates/multitriangle_exclusion.json` and the two clean replay transcripts.

Cycle/four-theta core exhaustion and support repairs. Primary: `src/core_enumerator.py`. Additional regeneration: `publication/src/regenerate_bounded_graph_universes.py`. Independent review: `publication/review/review_bounded_universes.py`. Output: `publication/certificates/bounded_universes.json`.

Five- and six-port theta directed atlases. Primary: `publication/src/regenerate_theta_atlases.py`. Independent review: `publication/review/review_directed_universes.py`. Outputs: the `canonical_theta_k5_*` and `canonical_theta_k6_*` certificates.

Cycle and cross-generator atlas. Primary: `publication/src/regenerate_cycle_atlas.py`. Independent review: `publication/review/review_bounded_universes.py`. Output: `canonical_cycle_cross_summary.json`.

Seven-port residual universe and strict separators. Primary: `src/verify_seven_port_closure.py`. Independent review: `review/review_seven_port_closure.py`. Outputs: `seven_port_closure.json` and `seven_port_adversarial_review.json`.

Pointwise cut theorem. Primary: `src/verify_frozen_cut_theorem.py` and the sparse publication replay. Independent review: `review/review_pointwise_cut.py`. Outputs: `pointwise_cut_certificate.json` and its adversarial-review JSON.

Root reduction. Primary: `src/verify_root_reduction.py`. Independent review: `review/review_root_reduction.py`. Outputs: `root_reduction.json` and its adversarial-review JSON.

Ordinary triangle redirection. Primary: the publication JC triangle checker. Independent review: a separately coded tensor replay. Output: `ordinary_triangle_T_certificate.json`.

Positive gluing and bounded-subdivision lift. Primary: `src/verify_positive_gluing.py` and `src/verify_bounded_support_lift.py`. Independent review: the final synthesis reviewer. Outputs: the corresponding JSON files and transcripts.

Theta sharpness. Primary: `src/verify_theta_sharpness.py`. Independent review: the publication sharpness reviewer. Output: `theta_sharpness_certificate.json`.

Final local-to-global synthesis and genericity. Primary: `src/build_sharp_boundary_certificate.py`. Independent review: `review/review_final_synthesis.py` and the publication scope review. Output: `final_theorem.json`.

Manuscript, bibliography, figures, and archive. Primary: `publication/verify_publication.py`. Independent review: `publication/review/review_publication_package.py`. Outputs: the clean quick/full transcripts, PDF preflight report, and SHA-256 manifest.

C.1 Exhaustiveness assertions

The primitive generator proves exhaustion in the following order.

- (i) Cyclomatic number at most two and biconnectivity reduce the undirected core to a cycle or theta graph.
- (ii) Two triangle cycles force theta path lengths (1, 2, 2); the direct rooting proof and independent finite census exclude this core from W_{TC} .
- (iii) Event placement and orientation enumerate every binary acyclic rooted realization of those cores.
- (iv) The local S_{TC} criterion and minimum-repair enumeration characterize every permitted port occupancy.
- (v) Every binary subdivision is an ordered word of ordinary port vertices on a directed core segment.
- (vi) The rigid-support theorem reduces arbitrary words to support plus one- and two-port probes.
- (vii) Canonical mixed-graph codes quotient exactly by labelled isomorphism and, in the observational atlas, by valid ordinary triangle redirection.
- (viii) Directed pair records are generated from the Cartesian product of canonical source and permitted target signature classes and stored under an ordered source–target key; duplicate

keys are rejected.

The independent programs use a different canonicalization and reconstruct the same code sets and orbit decompositions.

C.2 Release policy

The top-level verifier fails if a required source, certificate, review assertion, manuscript file, figure, transcript, or manifest entry is absent; if a generated object differs from its frozen release copy; or if the PDF preflight reports missing fonts, unresolved references, or malformed pages. The quick command checks the exact theorem certificates and publication integrity. The full command also regenerates the finite graph universes and runs every independent reviewer. The archived transcripts record the software versions, elapsed time, and peak resident memory.

References

- [1] Muhammad Ardiyansyah. “Distinguishing Level-2 Phylogenetic Networks Using Phylogenetic Invariants”. In: *arXiv preprint arXiv:2104.12479* (2021). DOI: [10.48550/arXiv.2104.12479](https://doi.org/10.48550/arXiv.2104.12479).
- [2] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*. 2nd ed. Vol. 10. Algorithms and Computation in Mathematics. Berlin: Springer, 2006. DOI: [10.1007/3-540-33099-2](https://doi.org/10.1007/3-540-33099-2).
- [3] Jacek Bochnak, Michel Coste, and Marie-Françoise Roy. *Real Algebraic Geometry*. Vol. 36. Ergebnisse der Mathematik und ihrer Grenzgebiete. Berlin: Springer, 1998. DOI: [10.1007/978-3-662-03718-8](https://doi.org/10.1007/978-3-662-03718-8).
- [4] Jari Brits, Niels Holtgreffe, Leo van Iersel, and Samuel Martin. “On Tree–Network Distinguishability and Full Identifiability of Phylogenetic Networks”. In: *arXiv preprint arXiv:2607.12919v2* (2026). Version dated July 30, 2026. DOI: [10.48550/arXiv.2607.12919](https://doi.org/10.48550/arXiv.2607.12919).
- [5] Shelby Cox, Elizabeth Gross, and Samuel Martin. “Group-Based Phylogenetic Models on 3-Sunlet Networks”. In: *Bulletin of Mathematical Biology* 87 (2025), p. 132. DOI: [10.1007/s11538-025-01506-1](https://doi.org/10.1007/s11538-025-01506-1).
- [6] Bryan Currie, Aviva K. Englander, Jose A. Esparza-Lozano, Elizabeth Gross, Max Hill, Colby Long, Devon Olds, Kawika O’Connor, Udani Ranasinghe, and Christin Sum. “Semialgebraic Conditions for Identifying Triangles in Phylogenetic Networks”. In: *arXiv preprint arXiv:2606.26673* (2026). DOI: [10.48550/arXiv.2606.26673](https://doi.org/10.48550/arXiv.2606.26673).
- [7] Aviva K. Englander, Martin Frohn, Elizabeth Gross, Niels Holtgreffe, Leo van Iersel, Mark Jones, and Seth Sullivant. “Identifiability of Phylogenetic Level-2 Networks under the Jukes–Cantor Model”. In: *bioRxiv* (2025). Version posted December 23, 2025. DOI: [10.1101/2025.04.18.649493](https://doi.org/10.1101/2025.04.18.649493).
- [8] Steven N. Evans and T. P. Speed. “Invariants of Some Probability Models Used in Phylogenetic Inference”. In: *The Annals of Statistics* 21.1 (1993), pp. 355–377. DOI: [10.1214/aos/1176349030](https://doi.org/10.1214/aos/1176349030).
- [9] Niels Holtgreffe, Katharina T. Huber, Leo van Iersel, Mark Jones, and Vincent Moulton. “Characterizing Semi-Directed Phylogenetic Networks and Their Multi-Rootable Variants”. In: *Theory in Biosciences* 145.1 (2026), p. 4. DOI: [10.1007/s12064-025-00453-8](https://doi.org/10.1007/s12064-025-00453-8).

- [10] Fabio Pardi and Celine Scornavacca. “Reconstructible Phylogenetic Networks: Do Not Distinguish the Indistinguishable”. In: *PLOS Computational Biology* 11.4 (2015), e1004135. DOI: [10.1371/journal.pcbi.1004135](https://doi.org/10.1371/journal.pcbi.1004135).
- [11] Bernd Sturmfels and Seth Sullivant. “Toric Ideals of Phylogenetic Invariants”. In: *Journal of Computational Biology* 12.2 (2005), pp. 204–228. DOI: [10.1089/cmb.2005.12.204](https://doi.org/10.1089/cmb.2005.12.204).
- [12] Seth Sullivant. *Algebraic Statistics*. Vol. 194. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 2018. DOI: [10.1090/gsm/194](https://doi.org/10.1090/gsm/194).

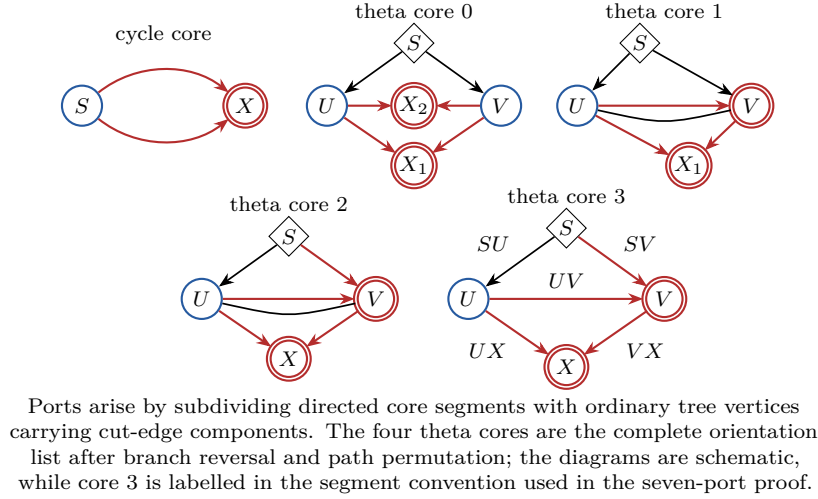


Figure 4: The cycle core and four oriented theta cores. The exact machine encodings, rather than the drawings, define the finite atlas.

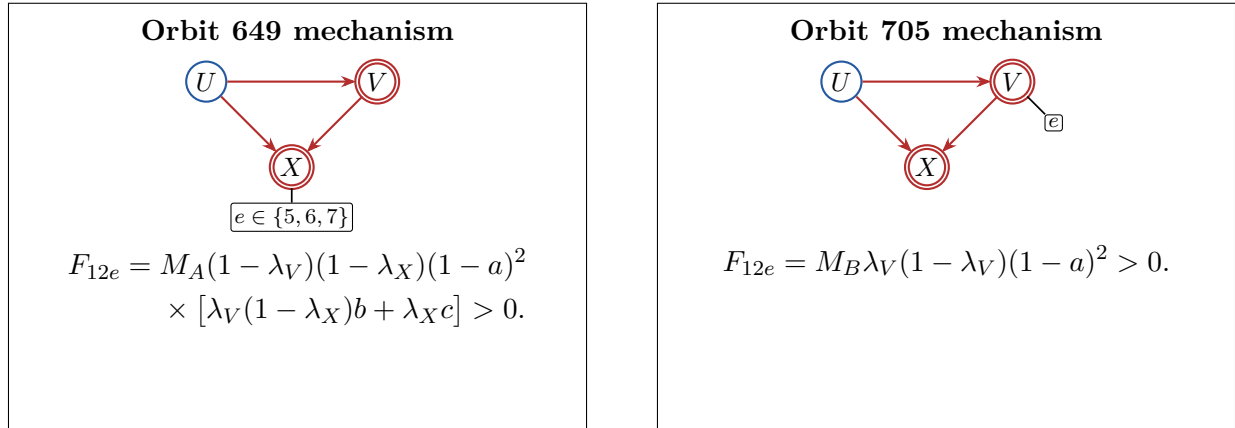


Figure 6: The two genuine seven-port residual mechanisms. In both panels the competing cycle has sink label 4, and hence $F_{12e} = 0$. The polynomial F_{abc} is the established JC trinet inequality; the new step is choosing a restored support label e that makes it a strict separator on the completed core-3 theta target.

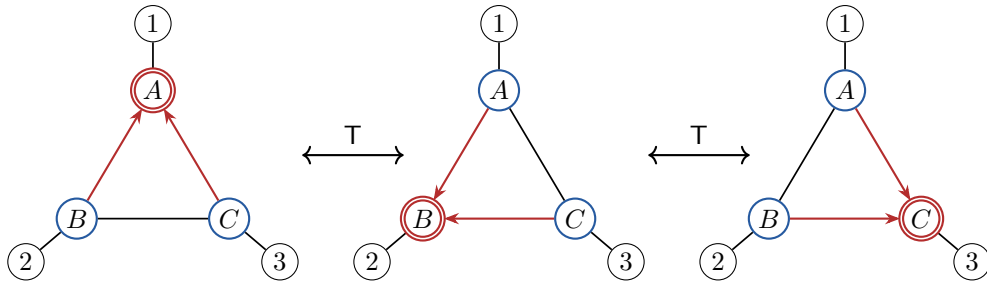
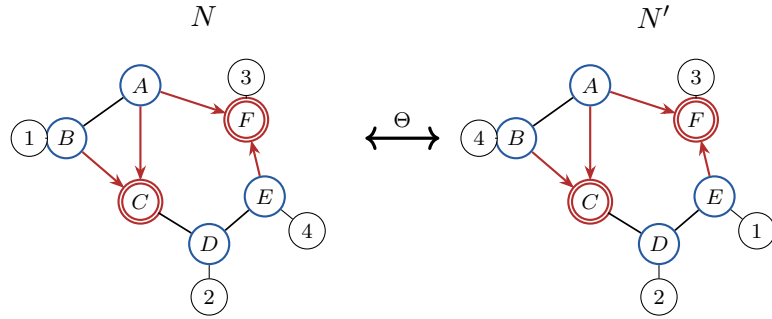
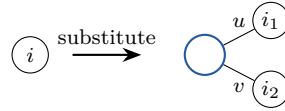


Figure 7: Ordinary triangle redirection T . The labelled underlying graph and all arrowheads outside the triangle are fixed; only the local reticulation designation changes. The three models share a common four-dimensional regular JC boundary-tensor germ, but equality of their complete open stochastic images is not asserted.



The pendant components at B and E are exchanged. The two standard semi-directed graphs are nonisomorphic and not $\mathbb{T}$ -equivalent. Each has five admissible rootings, exactly two tree-child, so both lie in $W_{\text{TC}} \setminus S_{\text{TC}}$.



$$\tilde{P}(g_X, g_1, g_2) = P(g_X, g_1 \oplus g_2) u^{\mathbf{1}_{[g_1 \neq 0]}} v^{\mathbf{1}_{[g_2 \neq 0]}}$$

Figure 8: The Theta sharpness pair and identical leaf substitution. Repeated substitution gives a non- $\mathbb{T}$ ambiguity in $W_{\text{TC}} \setminus S_{\text{TC}}$ for every $n \geq 4$.